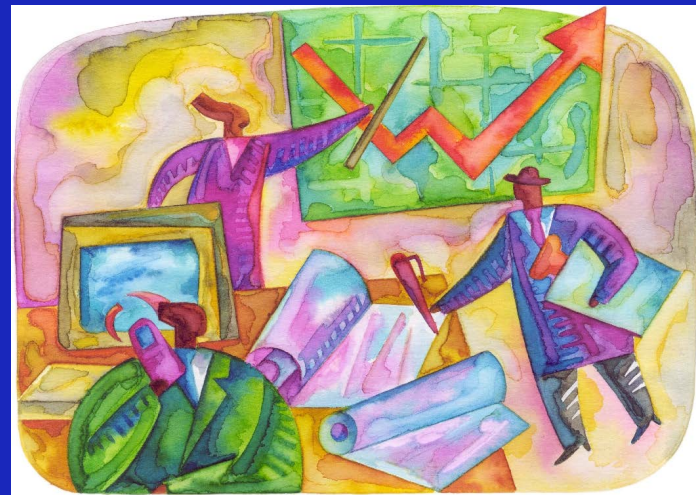


Scientific Poster Design

How to keep your poster from resembling an “abstract painting”



CCMR

Cornell Center for Materials Research

Cornell University, Ithaca, NY

<http://www.ccmr.cornell.edu>

A poster can be better than giving a talk

More efficient because:

- you totally bomb at giving talks
- can be viewed while you nap
- can hang in the department for years
- can reach folks not in your field of research

Posters serve as...

An advertisement of your hard work



Kool, wow!, check
this out!, you must
be smart!



It's just an illustrated abstract





Poster title goes here, containing strictly
only the essential number of words...

Author's Name/s Goes Here, Author's Name/s Goes Here, Author's Name/s Goes Here
Address/es Goes Here, Address/es Goes Here, Address/es Goes Here

Introduction

First ...

Check with conference organizers on their specifications of abstract and/or poster before you arrive at the poster session. Make the poster size and layout as professional as possible.

The poster should be 36 inches wide by 48 inches high (landscape format). Do not change the page size. You can scale the text and/or images to fit the poster. You must use a laser printer with a resolution of at least 600 dpi.

Be sure to leave a 1-inch margin all around the poster. Do not use a color background for the poster. Do not use a color background for the poster. Do not use a color background for the poster.

Method

Tip for making a successful poster ...

- Rewrite your paper in poster format. (Simply everything and use a laser printer.)
- Headings should be in bold. Do not use a color background for the poster. Do not use a color background for the poster. Do not use a color background for the poster.
- Use a laser printer with a resolution of at least 600 dpi.
- When laying out your poster, leave breathing space around the text. Do not crowd the poster.
- Try using photographs or diagrams to illustrate your points. Do not use a color background for the poster. Do not use a color background for the poster. Do not use a color background for the poster.
- (Tip: check and get someone to proofread.)

Results

Importing/Exporting files ...

Images such as photographs, graphs, diagrams, logos, etc. can be saved in the poster. ... then

To save space, use a laser printer with a resolution of at least 600 dpi. Do not use a color background for the poster. Do not use a color background for the poster. Do not use a color background for the poster.

Be sure to use a laser printer with a resolution of at least 600 dpi. Do not use a color background for the poster. Do not use a color background for the poster. Do not use a color background for the poster.

Do not use a color background for the poster. Do not use a color background for the poster. Do not use a color background for the poster.

Printing and Lamination

Once you have completed your poster, bring it to the VU to printing. (We will process it and deliver it to you.) Do not use a color background for the poster. Do not use a color background for the poster. Do not use a color background for the poster.

Be sure to use a laser printer with a resolution of at least 600 dpi. Do not use a color background for the poster. Do not use a color background for the poster. Do not use a color background for the poster.

Conclusion

For more information on
Poster Design, Scanning and Digital Photography,
and Image/Video.

Contact:
Medical Illustration Unit
Princeton Medical Center
Princeton, NJ 08540
Small: 609-255-2222, ext. 2222
Large: 609-255-2222, ext. 2222

Acknowledgements

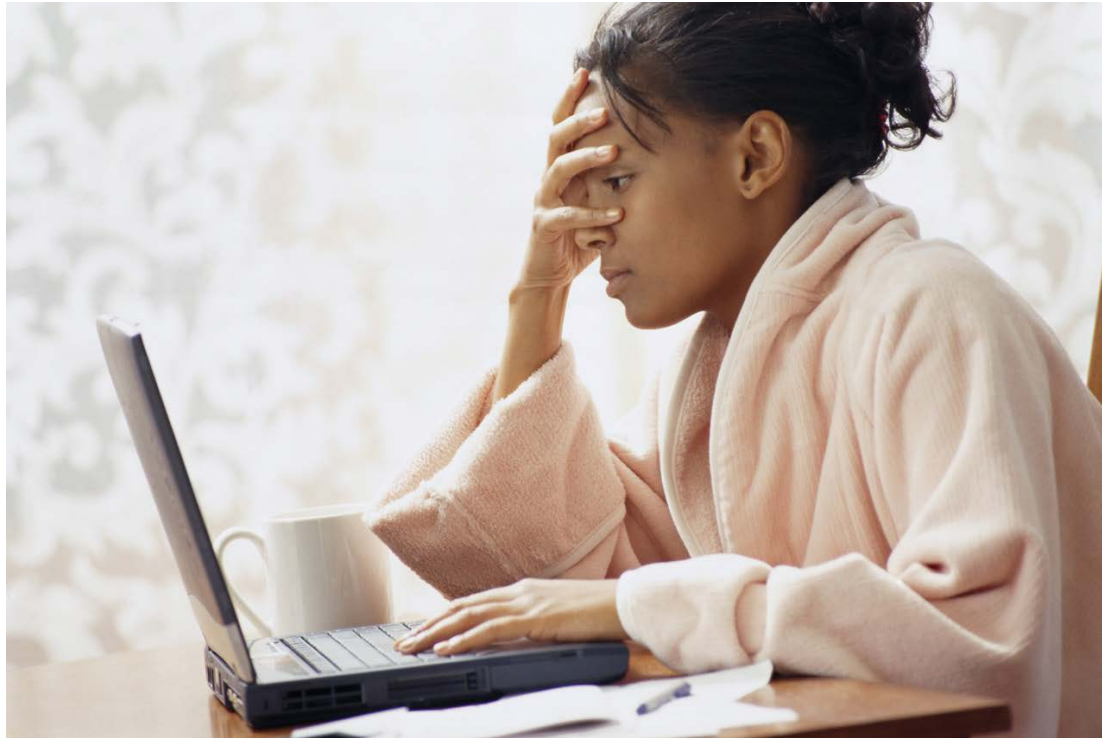
Just highlight the names of those who helped you.
Remember to thank them.

Is my abstract effective?

- Why should anyone care?
- What am I adding to current knowledge?
- Do I need to explain methods?
- Have I told them what I found and recommend?



A portrait of a grad student

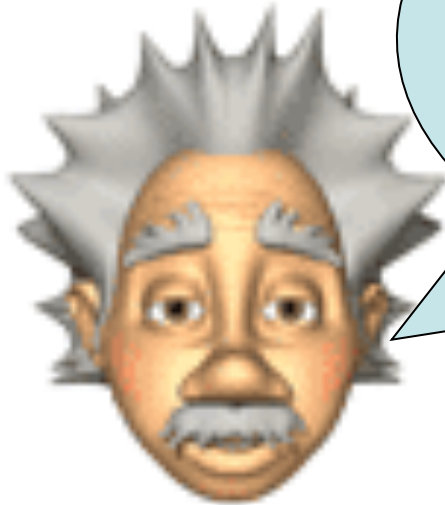


@#&%!@#\$, I have 12 hours to throw this thing together and get it printed before it's due.

How do I get months and years of research onto my poster?

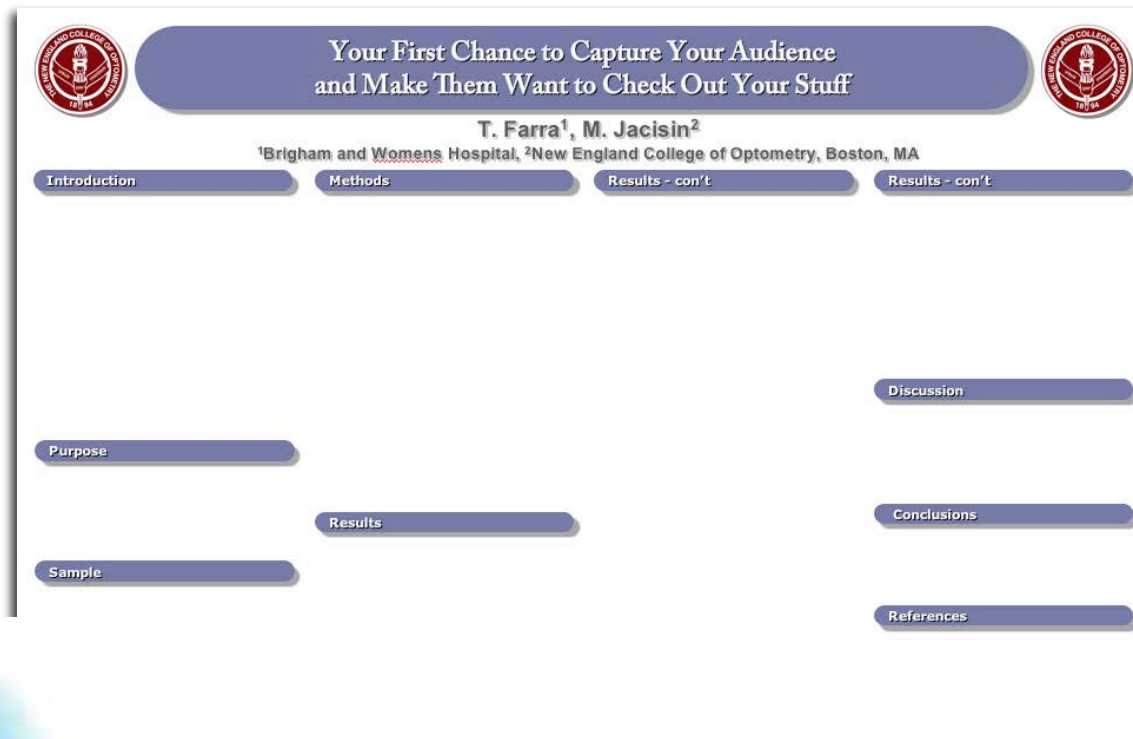


- Your poster is a short story
- Describe a few major points
- Arouse the reader's interest to read on
- Limit it to 250 words



Recite after me,
Less is best!

Simplify your paper into poster format



The poster template is a rectangular layout with a white background. At the top, there is a blue header bar with the Cornell University logo on the left and the NSF logo on the right. The title "Your First Chance to Capture Your Audience and Make Them Want to Check Out Your Stuff" is centered in the header. Below the title, the authors "T. Farra¹, M. Jacisin²" are listed, followed by their affiliations: "¹Brigham and Womens Hospital, ²New England College of Optometry, Boston, MA". The main body of the poster is divided into sections by blue rounded rectangles. The sections are: Introduction, Methods, Results - con't, Results - con't, Purpose, Results, Discussion, Conclusions, Sample, and References. The sections are arranged in a grid-like fashion, with some sections having more space than others.

 Your First Chance to Capture Your Audience
and Make Them Want to Check Out Your Stuff 

T. Farra¹, M. Jacisin²
¹Brigham and Womens Hospital, ²New England College of Optometry, Boston, MA

Introduction Methods Results - con't Results - con't

Purpose Results Discussion

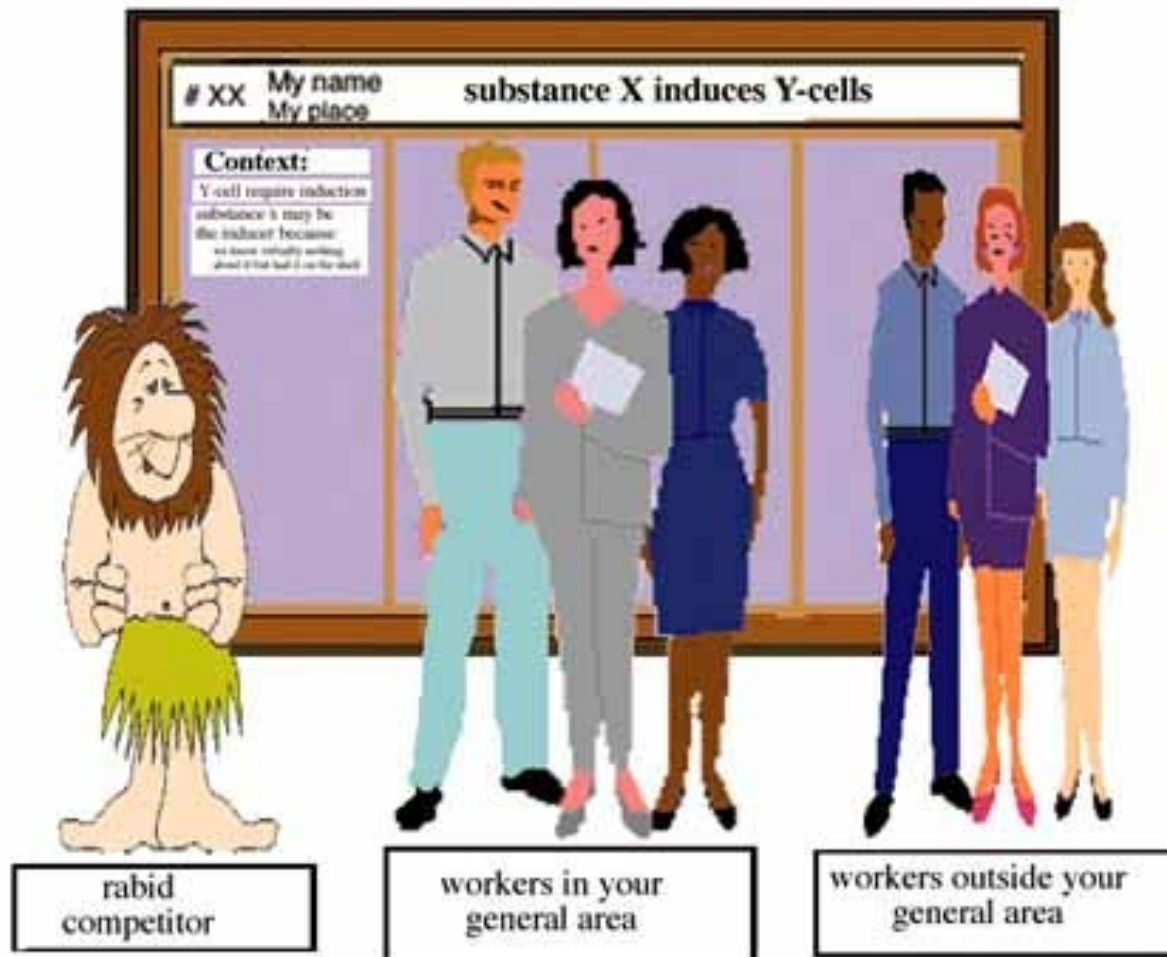
Sample Conclusions

References



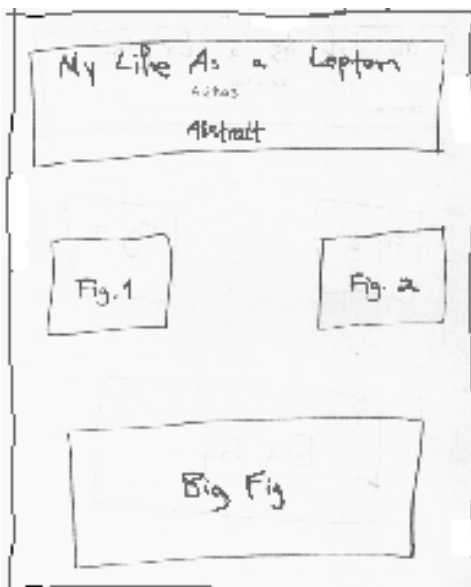
Find out the size required!

Who's my audience?



[illegible]

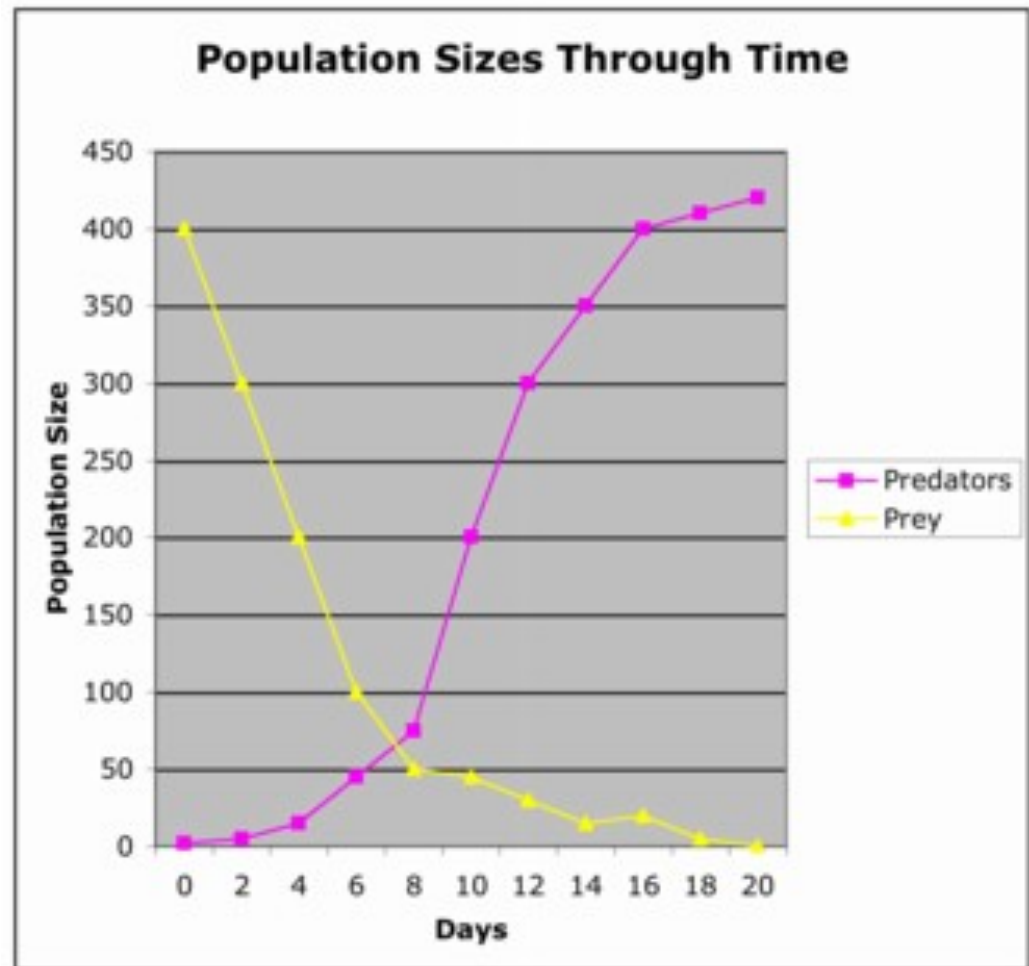
Carefully
omits
interpretations

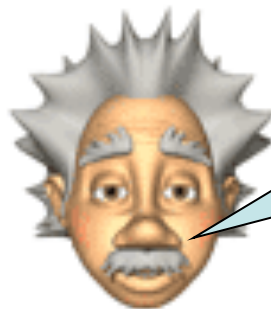
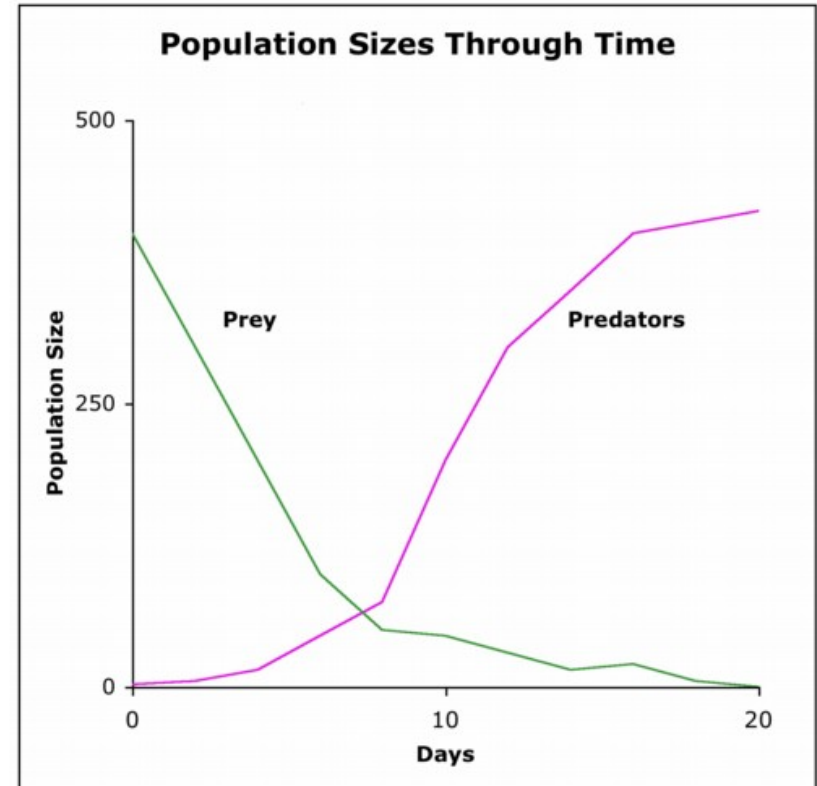
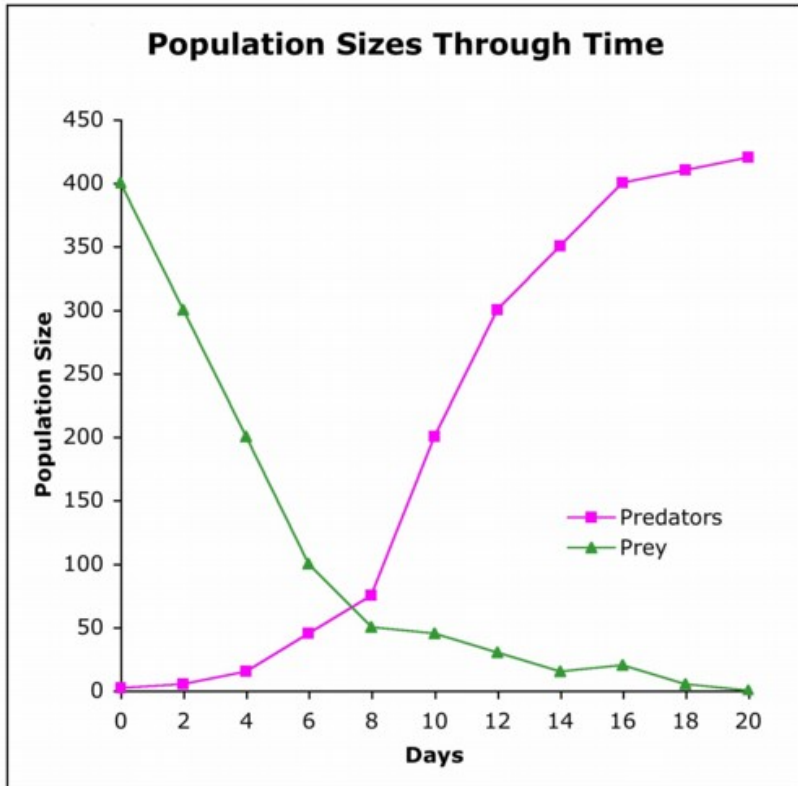


Start putting
together your
2 main elements

1) Simple, effective data displays

Don't make them stand on their heads to read your data!

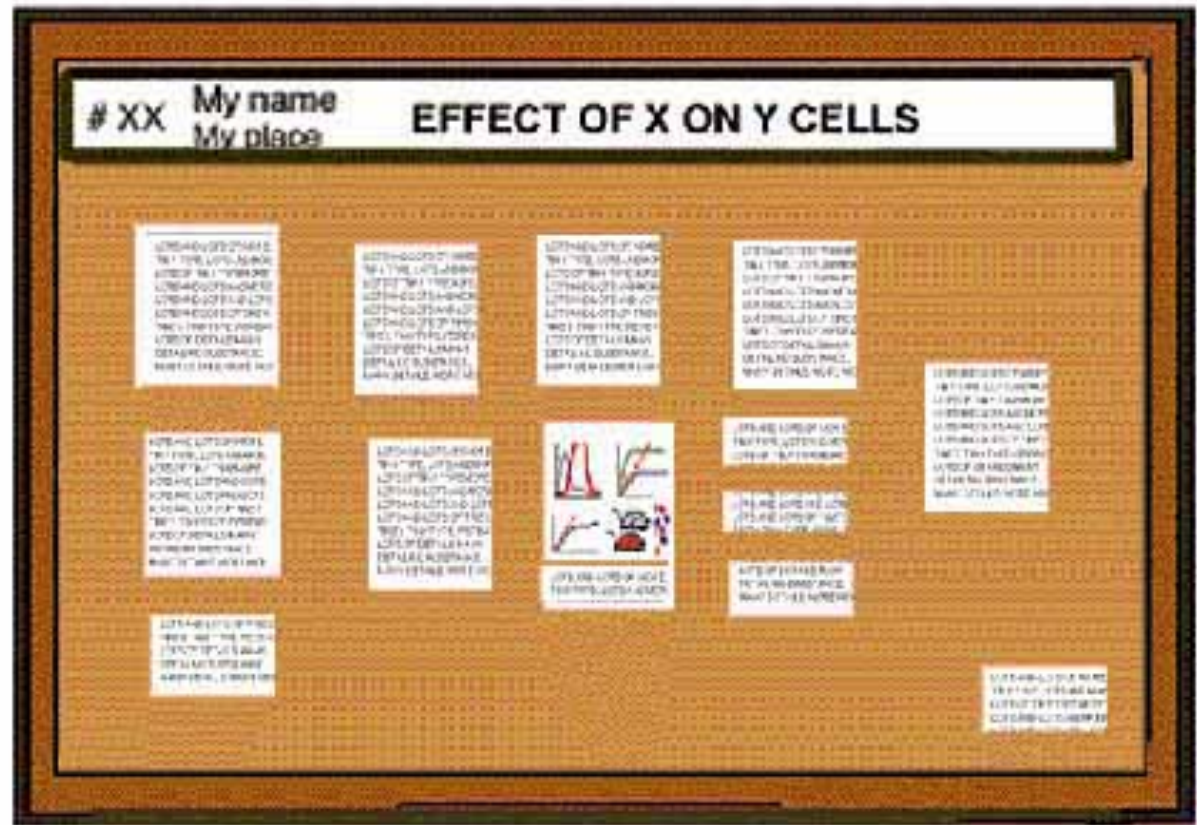




Keep it simple
but effective

2) Small blocks of supporting text

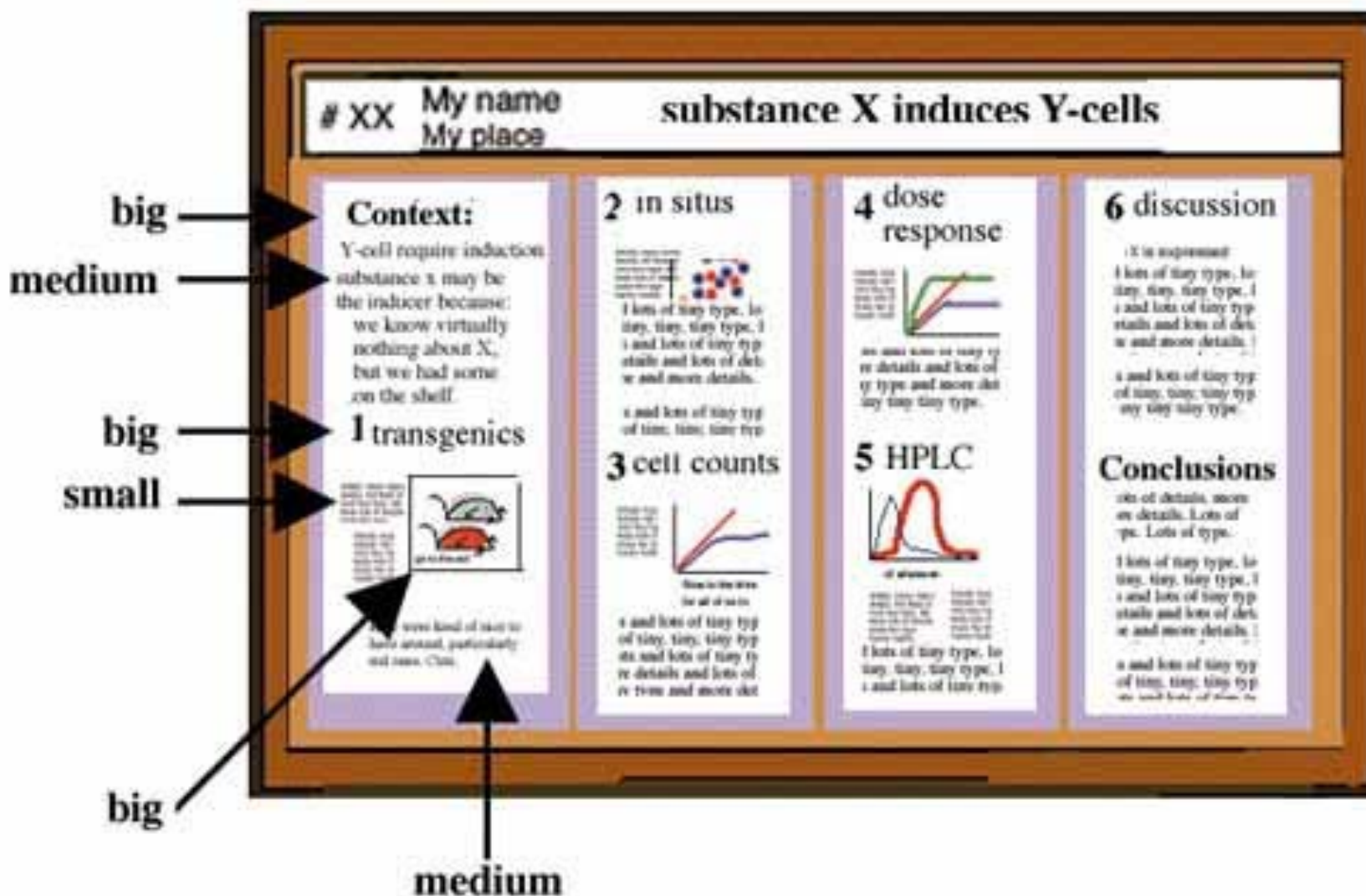
The need for chairs in front of your poster will not go over well

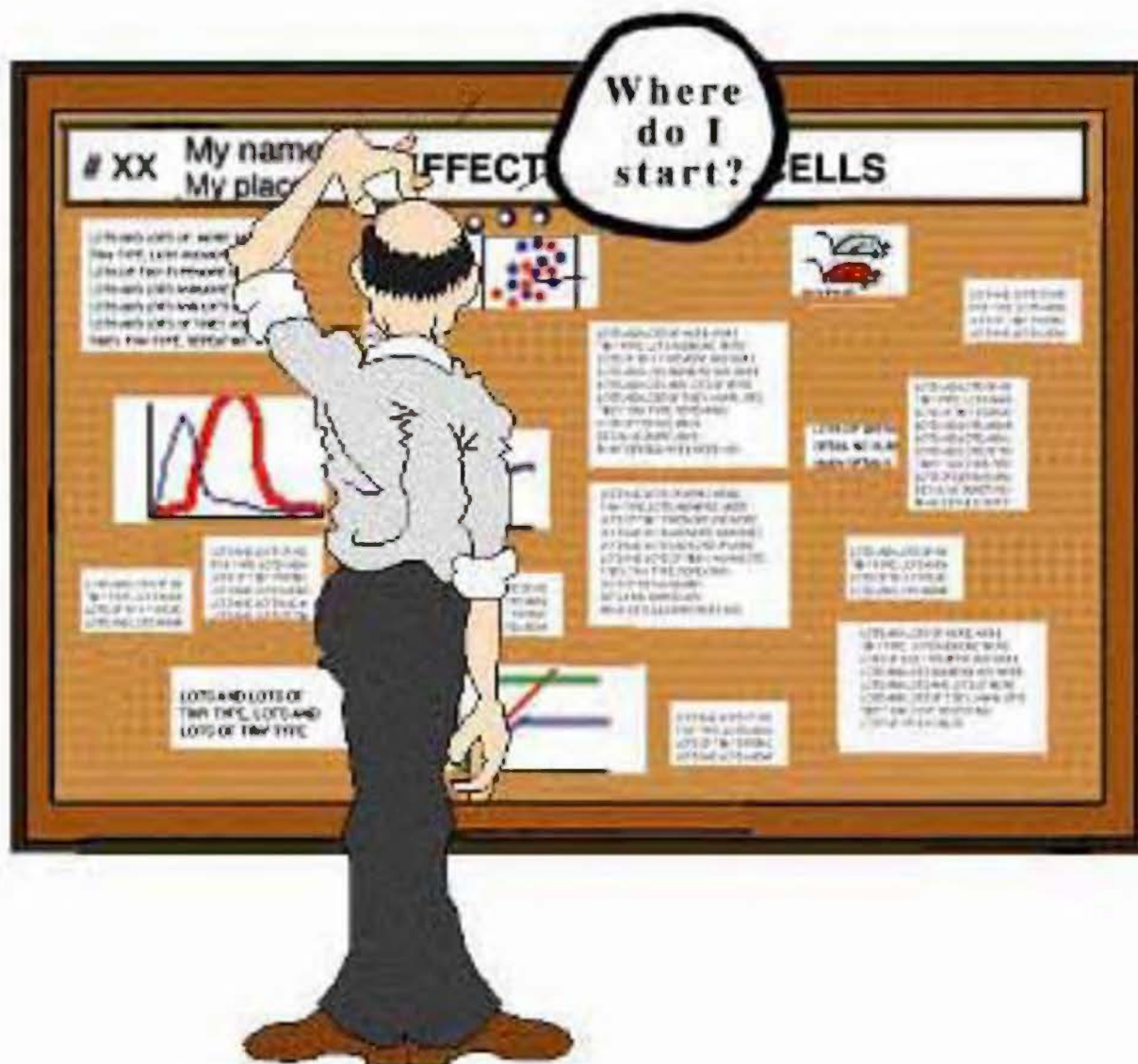


Your copy should answer...

# XX	My name My place	EFFECT OF X ON Y CELLS
Why?	Methods?	What do I recommend?
What am I adding?	What did I find?	

I could actually read this







Pick a software program

Although you'll probably gravitate towards PowerPoint,
consider a true design program.

PowerPoint



- OK, but the colors will fool you
- Easy to use
- Inflexible
- Designed for overhead projection

Adobe Illustrator or InDesign



- Excellent
- More difficult to learn
- What you see is what you get
- Others: Canvas, Publish-It, Corel Draw, LaTeX, etc.



Let's design a poster!

[illegible]

The Secrets of Readable Text:

**# XX My name
My place**

EFFECT OF X ON Y CELLS

INTRODUCTION

LOTS AND LOTS OF
THIS TYPE, LOTS AND
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RESULTS

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DOSE RESPONSE

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LOTS AND LOTS OF THIS TYPE

conclusions

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LOTS AND LOTS OF THIS TYPE
LOTS AND LOTS OF THIS TYPE

**Large type
states methods,
not results**

**Results
artfully buried in a
methods description**

**Carefully
omits
interpretations**

Poster title goes here, containing strictly only the essential number of words...

Author's Name/s Goes Here, Author's Name/s Goes Here
Address Goes Here, Address Goes Here

Introduction

Check all contents against the specifications of the presentation, including the following: machine-readable format, font size, and line spacing.

The presentation should be 10-12 pages long, including the title page and the last page. The presentation should be 10-12 pages long, including the title page and the last page.

Check all contents against the specifications of the presentation, including the following: machine-readable format, font size, and line spacing.

Abstract

Check all contents against the specifications of the presentation, including the following: machine-readable format, font size, and line spacing.

The presentation should be 10-12 pages long, including the title page and the last page. The presentation should be 10-12 pages long, including the title page and the last page.

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Method

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Results

Check all contents against the specifications of the presentation, including the following: machine-readable format, font size, and line spacing.

The presentation should be 10-12 pages long, including the title page and the last page. The presentation should be 10-12 pages long, including the title page and the last page.

Check all contents against the specifications of the presentation, including the following: machine-readable format, font size, and line spacing.

Conclusion

Check all contents against the specifications of the presentation, including the following: machine-readable format, font size, and line spacing.

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Acknowledgments

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The presentation should be 10-12 pages long, including the title page and the last page. The presentation should be 10-12 pages long, including the title page and the last page.

Check all contents against the specifications of the presentation, including the following: machine-readable format, font size, and line spacing.

- Leave breathing space around your text
 - Plain fonts even serif here
 - Same size and style
 - Left-aligned
- The reason is...



Hi there, my name is mitch collinsworth
and I would like to tell you about myself and how
I got this job at cornell. Well you see, my
uncle had a friend who knew my cousin on
the other side and his daughter worked for
facilities. I was down on my luck and my sister
told me she knew a guy who's
nephew's
wife's kid worked for this guys
father
and what can I say , he hired



Hi there, my name is mitch collinsworth and I would like to tell you about myself and how I got this job at cornell. Well you see, my uncle had a friend who knew my cousin on the other side and his daughter worked for facilities. I was down on my luck and my sister told me she knew a guy who's nephew's wife's kid worked for this guys father and what can I say, he hired me with no questions asked and just told me to keep my mouth shut. So here I am at CCMR.

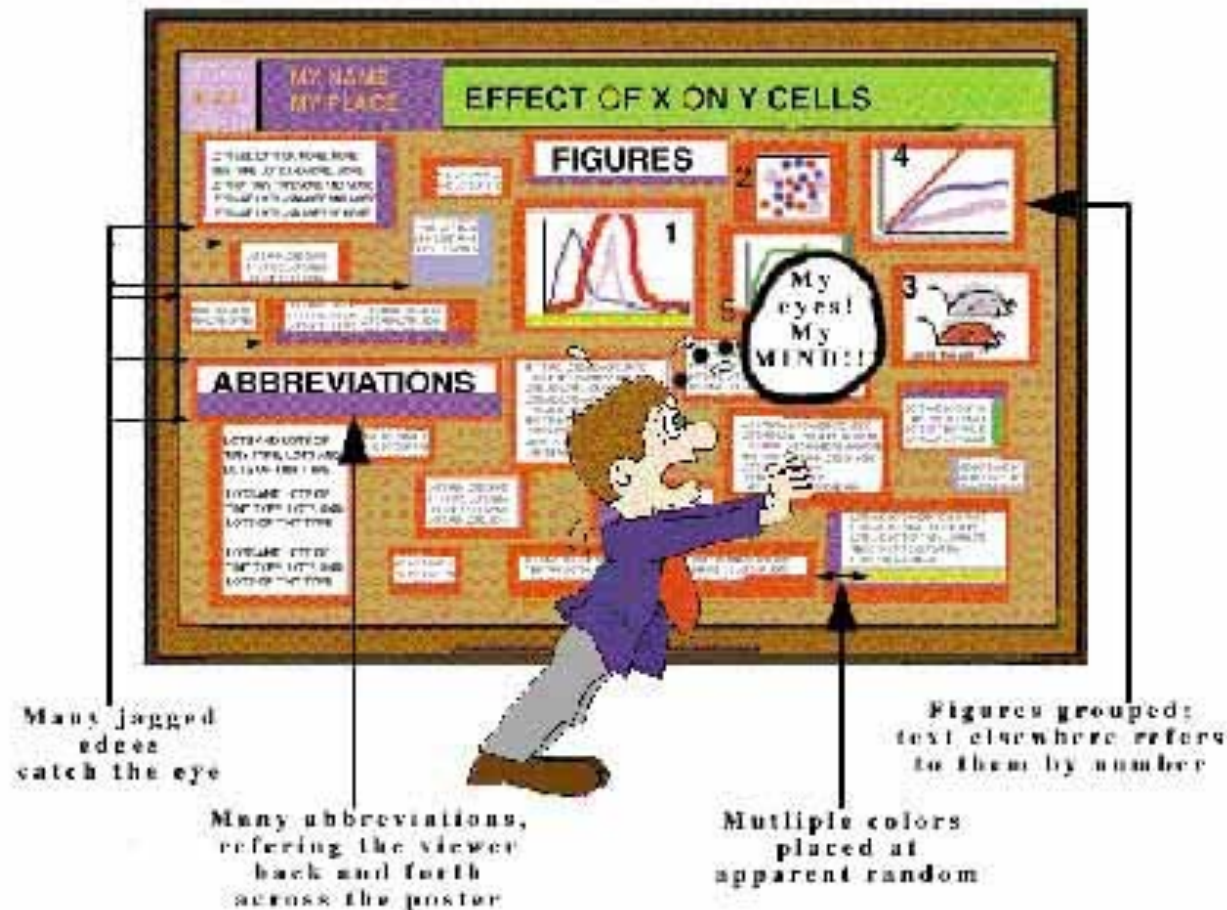


CCMR

Cornell Center for Materials Research

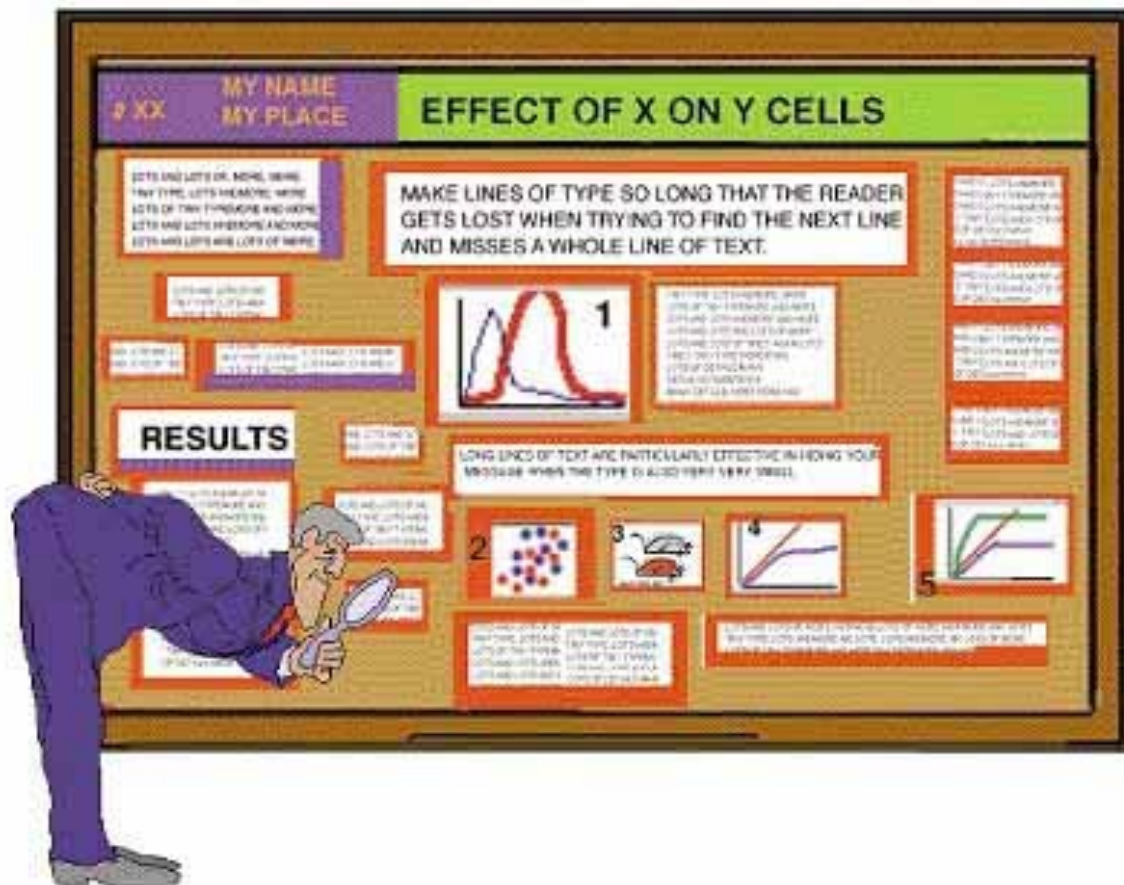


Easy for the eye to follow



Utter chaos will make folks dizzy!

Can anyone read your body text?



Text sizes:

Title: 85 point

Authors: 56pt

Sub-headings: 36pt

Body text: 24pt

Captions: 18pt

Karolinska Institutet

Your Ingenious Teaser Right Here to Woo Them Down to the Body

The name of the poster 24pt regular

Conclusions first: 44 pt bold

Always put the most important part - your conclusions - first! Place your conclusions in the upper left hand corner of your poster. Prepare your material from the reader's perspective. What was done, by who and your conclusion has to be understood within a couple of second's reading! Use active voice when writing the text. textsize: 34 pt regular

Introduction

Posters are primarily visual presentations. Your poster should be dominated by self-explanatory illustrations such as graphs and pictures while the amount of text should be kept to the minimum.

Your aim

Your poster is an advertisement for your research and as such it needs to be eye-catching and straight to the point. You only have seconds, or at best a few minutes to attract the attention of the visitor to a poster session. Keep your message short and clear

Your message

Keep your message clear and your text concise. Decide what is relevant for this poster and try to get your message across to your target group.

Layout, photos and print

Contact [Mediascan](#) at University Library for help with layout and image enhancement. For printouts and professional photographers contact [Björkstam](#). For more information: [www.bjorkstam.se](#)

Tips:

The best font for text blocks that are as short as they should be on a poster is a Sans Serif typeface family. Therefore, use sans serif fonts such as Arial or ~~Myriad~~ sans rather than serif fonts like Times or Courier. AVOID CAPITAL LETTERS IN TEXTS THAT ARE LONGER THAN ONE LINE, SINCE THEY ARE MORE DIFFICULT TO READ.

Handouts

If you succeed in getting the reader's attention, provide her/him with more detailed information in the form of handouts or printed articles. Include references on your handout instead of your poster.

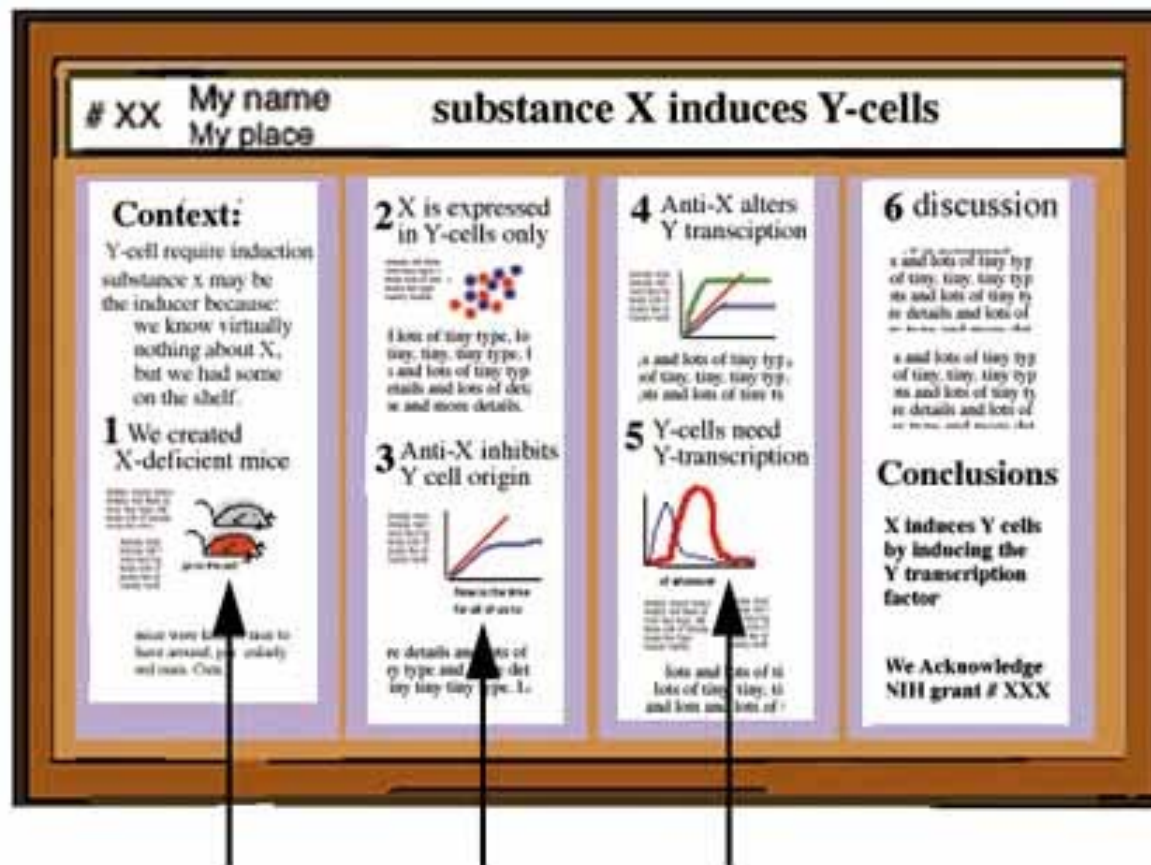
It is always nice to put in a picture and write some few short notes of what's going on in the future. Put handouts, business cards, nearby - on a table or in an envelope hung with the poster.

Use pictures or illustrations! Image caption 24pt regular

Always write a descriptive caption 24pt regular

Kontakt: Mediascan, University Library
Förtryck: Björkstam, Tryckeri
Telefon: 08 461 22 22
Fax: 08 461 22 22

Images and graphs say much more than words



BIG figures that use color

Keep posters visual!

BE STATE UNIVERSITY



Southern Flounder Exhibit Temperature-Dependent Sex Determination

J. Adam Luckenbach*, John Godwin and Russell Boeski
Department of Zoology, Box 7617, North Carolina State University, Raleigh, NC 27695



Introduction

Southern flounder (*Paralichthys lethostigma*) support valuable fisheries and show great promise for aquaculture. Female flounder are known to grow faster and reach larger adult sizes than males. Therefore, information on sex determination that might increase the ratio of female flounder is important for aquaculture.

Objective

This study was conducted to determine whether southern flounder exhibit temperature-dependent sex determination (TSD) and if growth is affected by rearing temperature.

Methods

- Southern flounder blood and skin were swabbed to collect eggs and sperm for *in vitro* fertilization.
- Fertilized larvae were reared from a natural diet on filter-sterilized to high protein pelleted feed and fed until satiation at least twice daily.
- Upon reaching a mean total length of 40 mm the juvenile flounder were stocked at equal densities into one of three temperatures (18, 23, or 28°C) for 245 days.
- Crustaceans were preserved and later sectioned at 2-6 microns.
- Sex-distinguishing markers were used to distinguish males (gonatogenesis) from females (oogenesis).

Histological Analysis

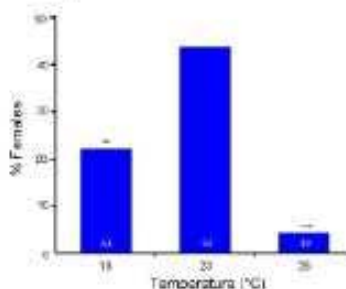


Male Differentiation



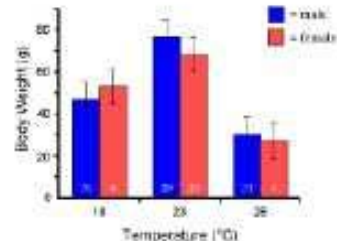
Female Differentiation

Temperature Affects Sex Determination



P < 0.01 and *P < 0.001 represent significant deviations from a 1:1 male:female sex ratio

Growth Does Not Differ by Sex



Results

- Sex was discernible in most fish greater than 120 mm long.
- High (28°C) temperatures produced 8% females.
- Low (18°C) temperatures produced 22% females.
- Mid-range (23°C) temperatures produced 44% females.
- Fish raised at high or low temperatures showed reduced growth compared to those at the mid-range temperature.
- Up to 245 days, no difference in growth existed between sexes.

Conclusions

- These findings indicate that sex determination in southern flounder is temperature-sensitive and temperature has a profound effect on growth.
- A mid-range rearing temperature (23°C) appears to maximize the number of females and promote best growth in young southern flounder.
- Although adult females are known to grow larger than males, no difference in growth between sexes occurred in age-0 to 1-year southern flounder.

Acknowledgements

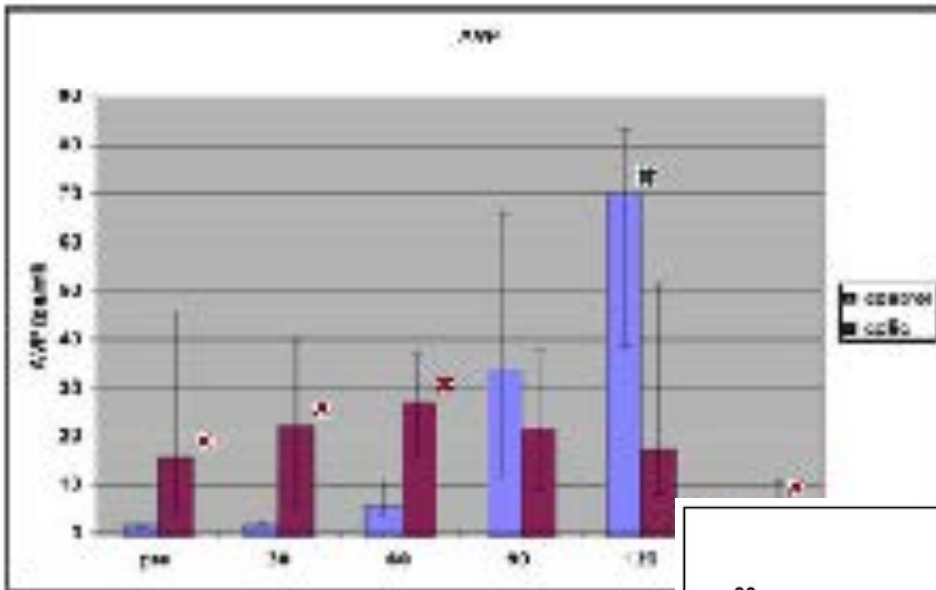
The authors acknowledge the Advanced Research Program of the National Marine Fisheries Service and the University of North Carolina Sea Grant College Program for funding this research. Special thanks to Lisa Wray and Beth Wray for help with the work.

Picture perfect photos

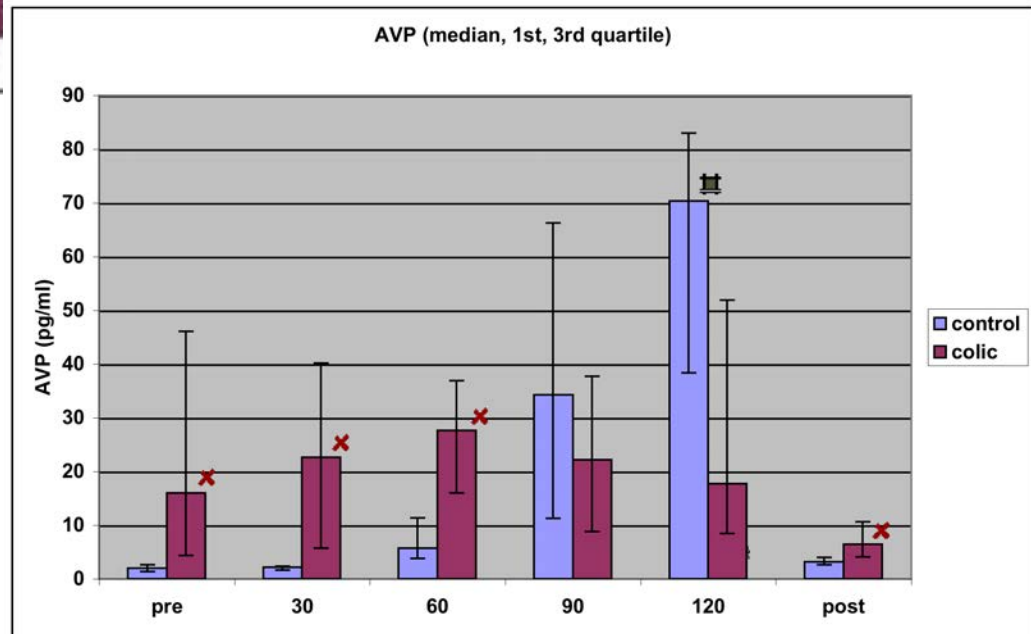
- Avoid resolution overkill!
At least 150 dpi, but no more than 300 dpi
- Save photos as jpg or png
Line art as a png (graphs)
- Web images are usually
poor resolution

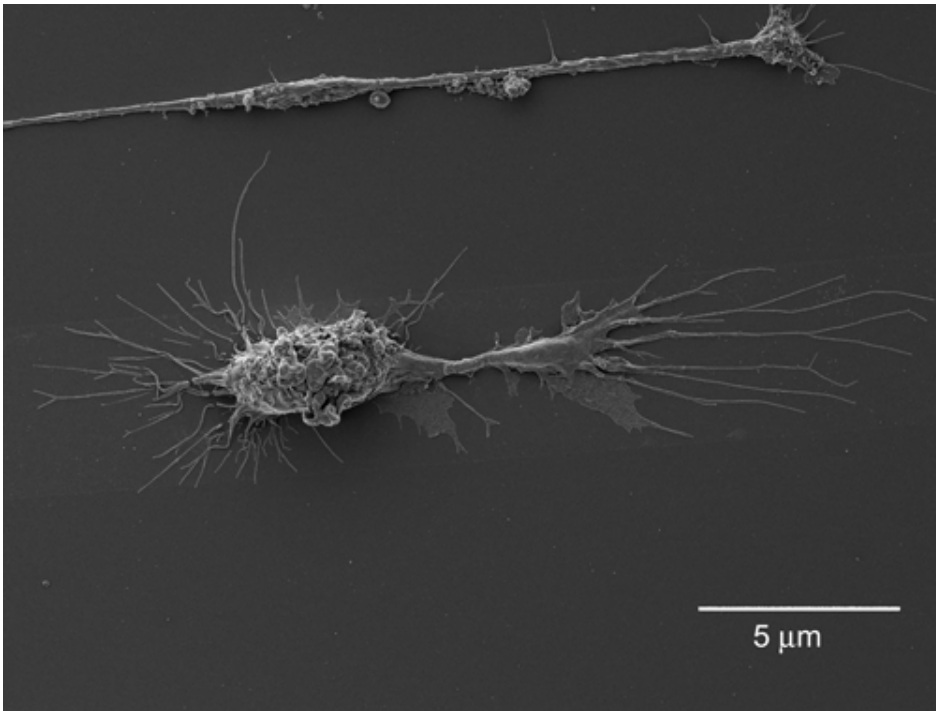


jpg



png

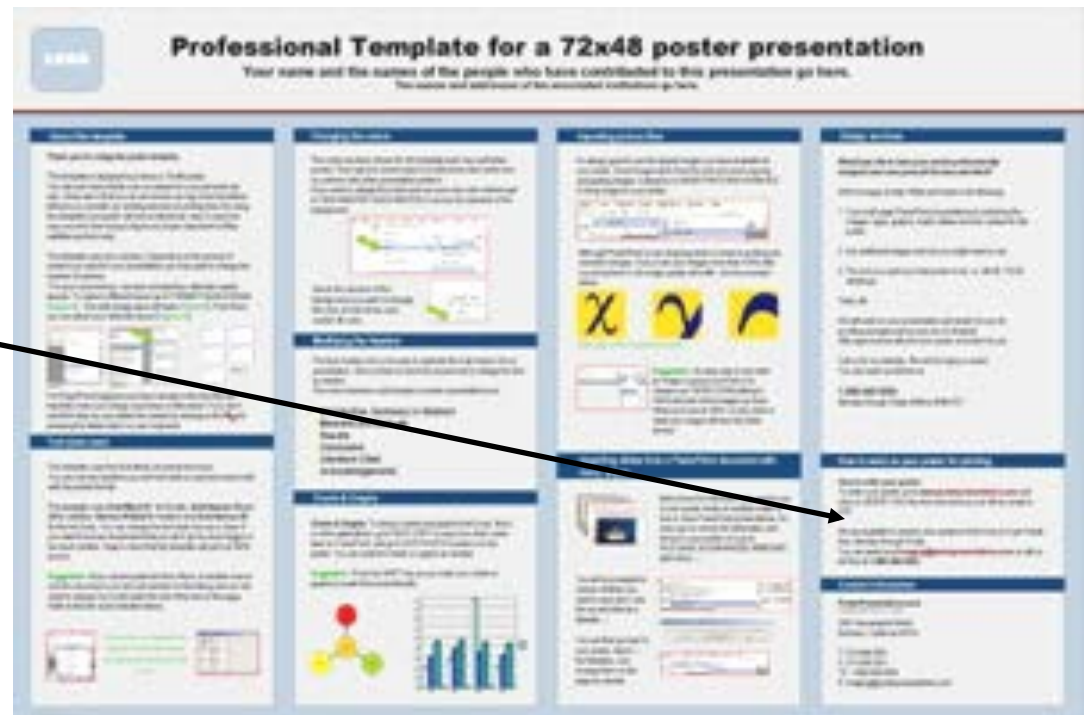




Your cool images
mean nothing
without a
scale bar or
description

Don't forget your funding acknowledgements

CNF-NSF-BMR, etc
Your department can
provide you with the
required wording



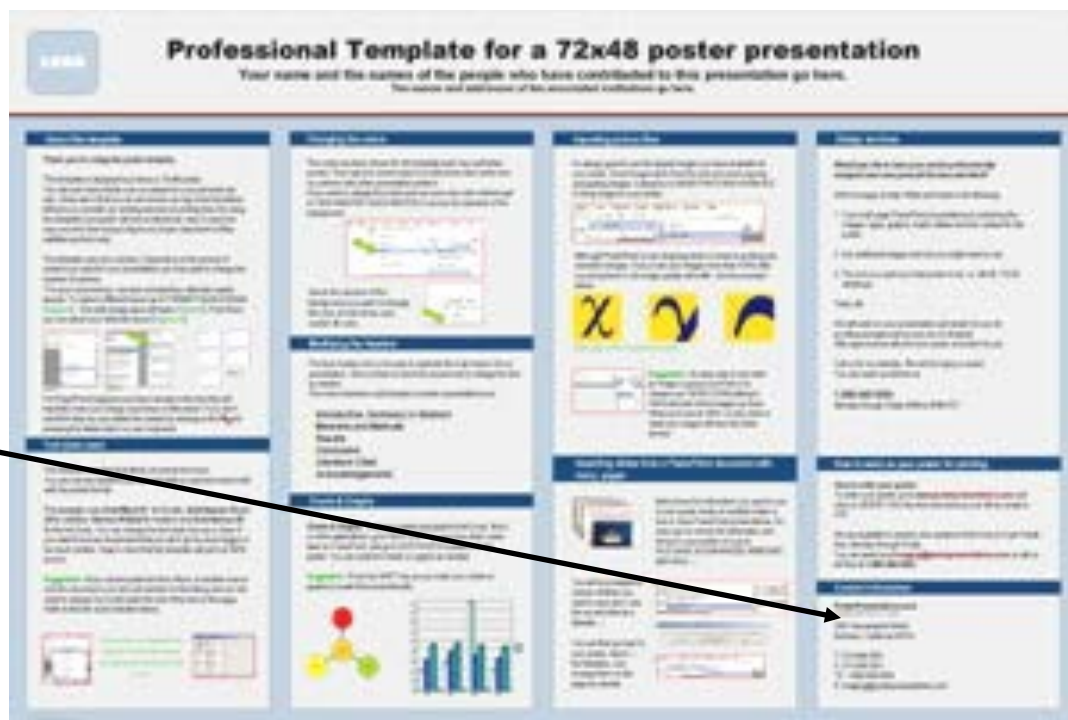
Your contact info!!!

Without it you'll become

“ya know, those guys with the awesome poster”

Include all
contact info:

- Mail address
- Phone
- E-mail



Using color to engage your readers

2-3 colors, no more!

Dark type on
light color background



Poster title goes here, containing strictly only the essential number of words...



Author's Name/s Goes Here, Author's Name/s Goes Here, Author's Name/s Goes Here

Address/es Goes Here, Address/es Goes Here, Address/es Goes Here

Introduction

Plot ...

Check with conference organizers on their specifications of abstracts and posters before you submit your poster. eg. make sure poster abstracts are no longer than 100 words.

The poster should be 36 inches wide (30 inches high), landscape (horizontal) format. Do not change the page size. You can scale to a smaller or larger size when printing. You need a different aspect ratio than a portrait vertical or a landscape poster template.

Use 12 point font. Do not use bold or italics. Do not use color. Do not use a different font size. Do not use a different font style. Do not use a different font color. Do not use a different font size. Do not use a different font style. Do not use a different font color.

Method

Tips for making a successful poster ...

- Read your paper in poster format. eg. ...
- Readings of more than 6 words should be bold and lower case, not all caps.
- Use a word processing program to create your poster. Use a word processing program to create your poster. Use a word processing program to create your poster.
- When laying out your poster, leave breathing space around your text. Don't crowd your poster.
- Try using photographs or colored graphics. Avoid long numerical tables.
- Spill check and get someone to proofread.

Figure 1: A typical poster layout. The figure is a 12 inch wide by 18 inch high. The figure is a 12 inch wide by 18 inch high. The figure is a 12 inch wide by 18 inch high.

Results

Importing the results ...

Images such as photographs, graphs, diagrams, logos, etc. can be added to the poster.

To import images into your poster, go through the menu: Insert > Picture > From File. The image will be imported into your poster. The image will be imported into your poster. The image will be imported into your poster.

Be aware of the image size. The average color photo (13.5 inch x 18 inch) would be about 3MB. (Note: 1MB is 1,000,000 bytes.) Call the University of Illinois at Urbana-Champaign for more information.

Do not use images from the web.

Figure 2: A typical poster layout. The figure is a 12 inch wide by 18 inch high. The figure is a 12 inch wide by 18 inch high. The figure is a 12 inch wide by 18 inch high.

Printing and Lamination

Once you have completed your poster, bring it to the University of Illinois at Urbana-Champaign for printing. We will provide a 12 inch x 18 inch poster. The poster will be printed and laminated.

Do not (Do not) use your poster until it has been printed. Allow at least 24 hours for the poster to be printed. Allow at least 24 hours for the poster to be printed.

Simply highlight the text to replace.

Cost ...

For poster printing and lamination charges contact the University of Illinois at Urbana-Champaign.

Aim

How to create a poster template ...

Simply highlight the text to replace it with your own. You can also use the University of Illinois at Urbana-Champaign poster template. The poster template is a 12 inch wide by 18 inch high. The poster template is a 12 inch wide by 18 inch high.

The poster template is a 12 inch wide by 18 inch high. The poster template is a 12 inch wide by 18 inch high. The poster template is a 12 inch wide by 18 inch high.

The poster template is a 12 inch wide by 18 inch high. The poster template is a 12 inch wide by 18 inch high. The poster template is a 12 inch wide by 18 inch high.

Conclusion

For more information on poster design, scanning and digital photography, and image resolution, contact the University of Illinois at Urbana-Champaign.

Contact:

Medical Illustration Unit
Phone: 248-2800
Email: medillustr@uiuc.edu
Web: <http://www.medillustr@uiuc.edu>

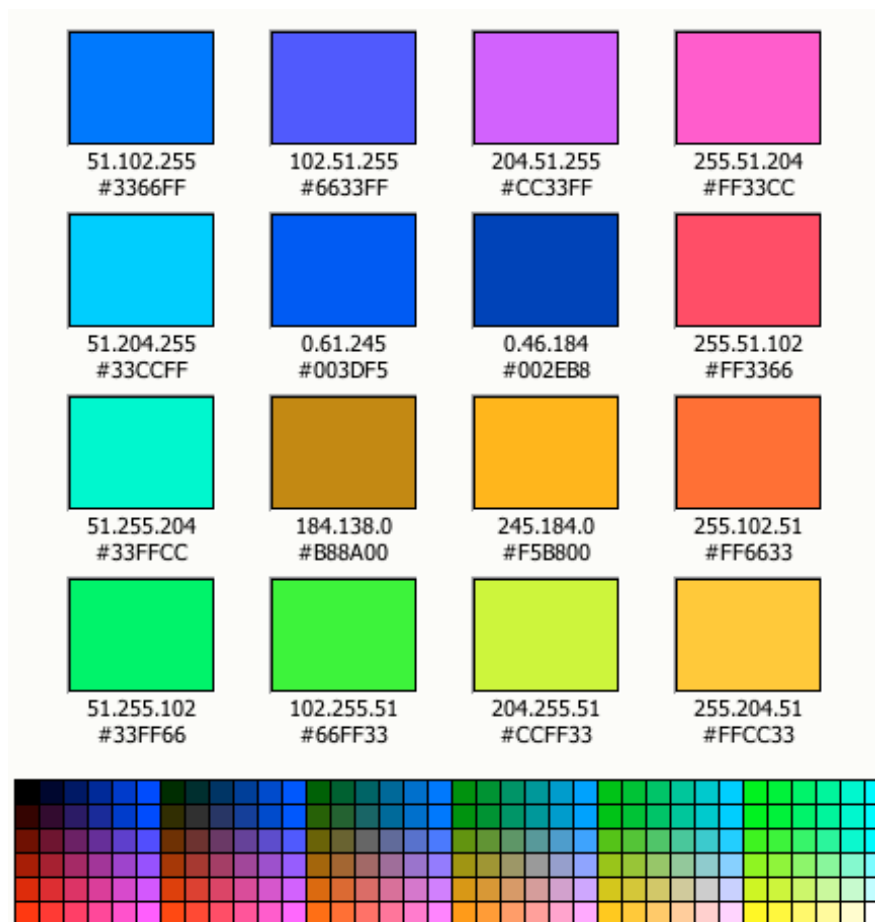
Acknowledgements

Just highlight the text to replace it with your own. Replace it with your own.

[illegible]

This attracts attention but tires out the eye

Be careful with the primary colors





Blue on Red appears blurry to the human eye.

Yellow on white is hard to read

Red on Blue appears blurry to the human eye.



• aeiko



• Peach Green & Seeds



• Rust

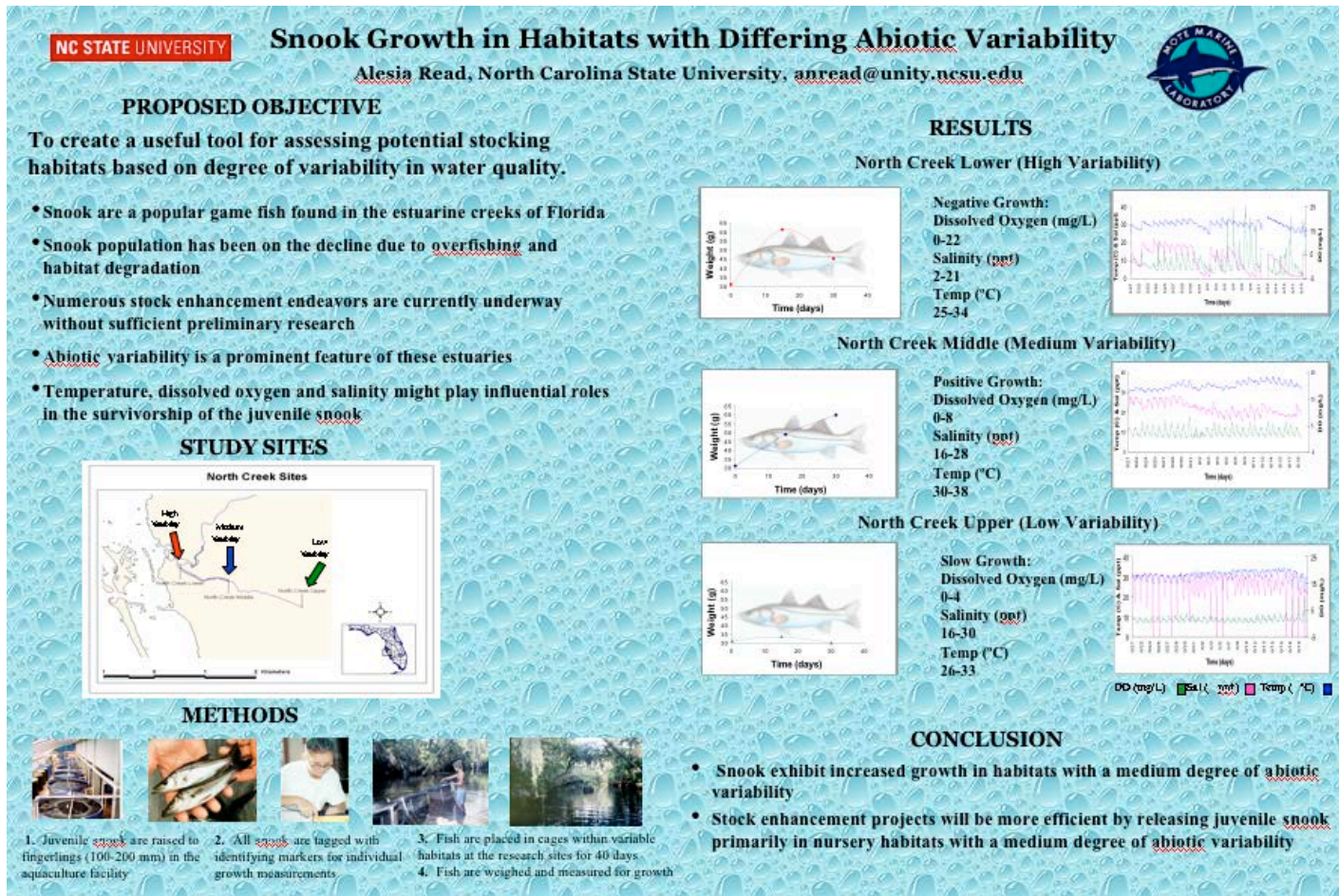


• dollar



<http://www.colorschemer.com/online.html>

Be aware of busy backgrounds





Southern Flounder Exhibit Temperature-Dependent Sex Determination

J. Adam Luckenbach*, John Godwin and Russell Boeski

Department of Zoology, Box 7617, North Carolina State University, Raleigh, NC 27695



Introduction

Southern flounder (*Paralichthys lethostigma*) support variable behavior and show great promise for aquaculture. Female flounder are known to grow faster and reach larger adult sizes than males. Therefore, information on sex determination may help increase the ratio of female flounder in aquaculture.

Objective

This study was conducted to determine whether southern flounder exhibit temperature-dependent sex determination (TSD), and if growth is affected by rearing temperature.

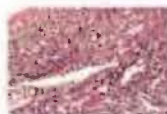
Methods

- Southern flounder broodstock were strip-spawned to collect eggs and sperm for *in vitro* fertilization.
- Fertilized larvae were reared from a natural diet (rotifers *Artemia*) to high protein (protein-based food) and fed until saturation in larval rearing (20%).
- Upon reaching a mean total length of 40 mm, the juvenile flounder were stocked at equal densities into one of three temperature (18, 23, or 28°C) for 245 days.
- Flounder were preserved and later sectioned at 2- μ m sections.
- Sex-distinguishing markers were used to distinguish males (gonatogenesis) from females (gonadosis).

Histological Analysis

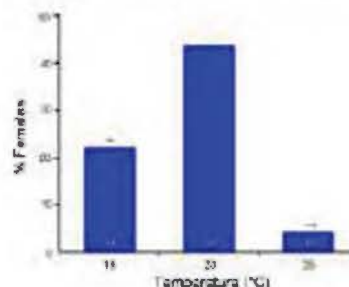


Male Differentiation



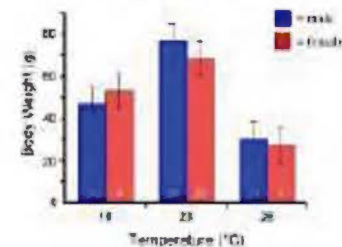
Female Differentiation

Temperature Affects Sex Determination



P < 0.01 and *P < 0.001 represent significant difference from 1:1 male:female sex ratio.

Growth Does Not Differ by Sex



Results

- Sex was discernible in most fish greater than 120 mm long.
- High (28°C) temperatures produced 5% females.
- Low (18°C) temperature produced 22% females.
- Mid-range (23°C) temperature produced 48% females.
- Flounder at high or low temperatures showed reduced growth compared to those at the mid-range temperature.
- Up to 245 days, no difference in growth existed between sexes.

Conclusions

- These findings indicate that sex determination in southern flounder is temperature-sensitive and temperature has a profound effect on growth.
- A mid-range rearing temperature (23°C) appears to maximize the number of females and promote better growth in young southern flounder.
- Although adult females are known to grow larger than males, no difference in growth between sexes occurred in age-0 to 1 year southern flounder.

Acknowledgements

The authors acknowledge the following: Kenneth Johnson of the National Marine Fisheries Service and the University of North Carolina Sea Grant College Program for funding for this study. Thanks to the following individuals for their assistance in completing this project: Dr. Russell Boeski.

The authors acknowledge the Financial Support of the National Science Foundation and the University of North Carolina Ben-Gurion College. Financial support for the research is gratefully acknowledged by the authors.

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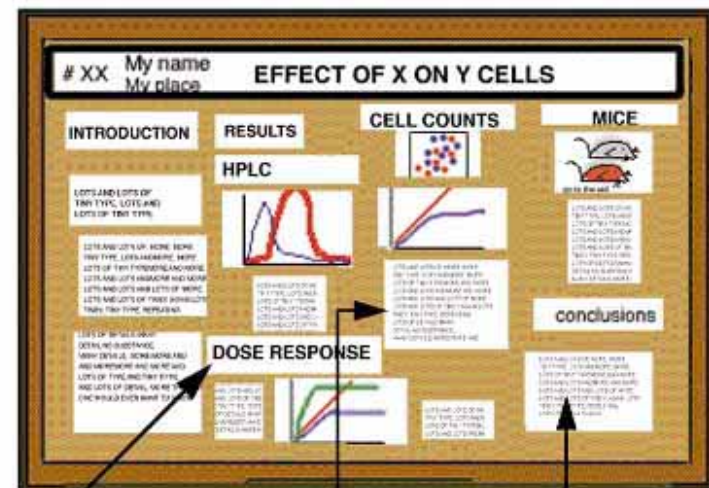
Print out a letter size draft

Can you read the type?

Are these the colors you really want?

Does it look too busy?

Do my main points pop?



Large type
states methods,
not results

Results
artfully buried in a
methods description

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interpretations

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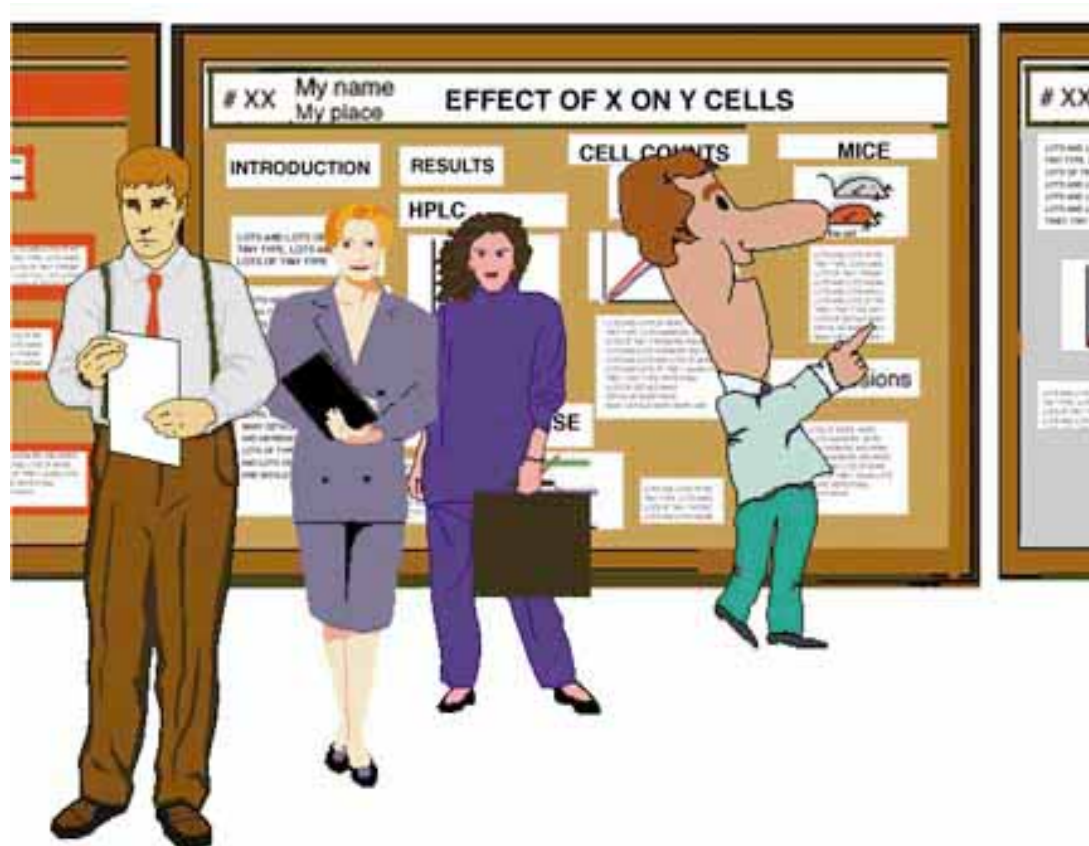
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Prepare a 3-5 minute verbal explanation

Is he ever
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- Provides a written record for interested folks
- Makes you look together
- Be sure to include complete contact information
- Might even get you a job!



Let's judge some designs
and see what you've learned

Using a Windbreak Habitat Model Across Broad Landscapes: The Effect of Local Landscape Composition and Geographic Location

George Hess¹, John Poulsen², Raymond O'Connor³, Jeff Bay³

1. Windbreaks as Habitat

Agricultural lands — fields, pastures, and orchards — are managed to produce food and fiber for people. In the U.S. Great Plains, an extensive agricultural landscape, windbreaks have been planted to protect fields, crops, livestock, and livestock from the prevailing wind. Windbreaks provide some of the most needed habitat for birds and other wildlife that people have come to value. Windbreaks make up about 25% of the wooded cover in Nebraska, much of the other wooded cover occurring along riparian corridors.

Although they protect soil from wind erosion and provide habitat for some species, windbreaks also contribute to the fragmentation of prairie grasslands. Prairie fragmentation negatively impacts prairie wildlife such as prairie prairie chickens, upland sandpeps, and prairie dogs.

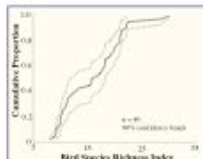
- Early windbreaks were planted using tree species common to the region and were established by local farmers.
- Most early windbreaks were planted in the 1930s and 1940s.
- Habitat characteristics of each windbreak were measured in 1994.
- Thirty-five farmers allowed windbreaks to mature in 1994.



2. Regional Evaluation of Windbreaks

The Environmental Monitoring and Assessment Program's Agricultural Land Use Group — charged with assessing the ecological condition of U.S. agricultural lands — conducted a pilot study to evaluate the habitat value of windbreaks on a regional basis. We decided to use a bird species richness index to measure the habitat value of individual windbreaks.

We selected a random sample of 80 windbreaks in Nebraska, based on a screening question on a USDA National Agricultural Statistics Service agricultural survey. In July 1994, field crews measured attributes of 80 windbreaks (see list of the farmers referred to participants). The data were used to estimate the value of windbreaks in breeding bird habitat in Nebraska.



3. Bird Species Richness Index

We used the U.S. Fish and Wildlife Service's Bird Species Richness Index (BSRI), which estimates the number of breeding bird species a single windbreak can support based on four windbreak attributes.

- Area has the greatest impact on bird diversity; larger windbreaks support more species. Area was measured by calculating perimeter.
- Height: Taller windbreaks provide more niches. Height was measured by photographic analysis.
- Vertical Structure: A more structurally complex windbreak provides more habitat niches measured by point sampling.
- Shape: Irregular shapes provide another habitat niche. Shape was measured.

- Using regression factors associated with each sample, we estimated the habitat value of windbreaks for the region (graph left).

- We estimated that half of Nebraska's windbreaks support fewer than 10 breeding bird species (graph left).

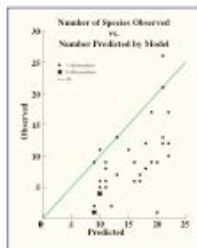
- We also estimated that between 87% and 98% of windbreaks are smaller than 1.5 hectares (data not shown), suggesting that few Nebraska windbreaks provide habitat for forest interior or area sensitive birds.

4. Validating BSRI Model

In 1994, a team of three ornithologists revisited 35 of the 80 windbreaks 10 years after their initial visit in 1984 and 1994.

Each windbreak was visited two times. Data were collected between one full hour before and one hour after sunrise. All observed birds were identified and recorded using spot mapping techniques. Two ornithologists, one of the authors, traveled with the team to the field and through the windbreak for each visit.

Because the windbreaks were mature, we assumed all species were detected.



5. Results of Validation

The model overestimates the number of bird species in the Nebraska windbreaks (graph left). However, the relative qualitative ranking of windbreaks is generally preserved. A total of 30 species were observed.

For strong, significant relationships were found between deviation of observed from predicted number of species and any windbreak attributes or the geographic location of individual windbreaks.

Forest interior, area-sensitive, and forest edge species occurred in the larger, taller, more complex windbreaks.

Openland and grassland species occurred in the smaller, shorter, less complex windbreaks.

6. Failure of the Model

There are several possible explanations for the failure of the model to predict accurately the number of birds species in the windbreaks.

1) Geographic differences in species richness. The model was developed in Kansas, which has 5-20 more species of bird than Nebraska. Breeding bird surveys species (Nebraska map of North America).

2) Dependence on different windbreak characteristics. The number of species in Nebraska's windbreaks depends differently on windbreak characteristics than did for number of species in Kansas.

3) Dependence on landscape-scale characteristics. The number of species in Nebraska's windbreaks depends on characteristics of the surrounding landscape.

7. Local Landscape-Scale Effects

Land cover data were collected for the quarter-section (360 acres) of the containing the sample windbreak. Cover categories were: tree, wooded, crop, grass/forbushes, forest, non-forested, and water. Forest and water were also recorded present (absent).

Landscape metrics computed included relative cover distributions, total edge length, edge:area ratio, number of patches, mean patch size, mean perimeter per patch, and size of largest field.

The relation between observed and predicted number of species was not significantly related to any of the landscape metrics. This suggests that neither a region's number of species using a windbreak depends primarily on landscape attributes.

8. Conclusions

1) The Bird Species Richness Index for windbreaks cannot be considered simple to describe species richness at large regional scales without either a carefully selected or a well-defined set of attributes for the landscape in a regional species pool.

2) Local landscape-scale composition and structure do not explain the failure of the model.

3) The presence of species pools in windbreaks is a function of landscape-scale characteristics. The model may be more useful for predicting the presence or absence of species pools than for predicting the total number of species present.

Acknowledgments: This work could not have been done without the many dedicated people at the National Agricultural Statistics Service who helped place and maintain the 1994 data collection effort, the field crews who collected the data, the Nebraska Department of Game and Parks who provided the Nebraska Breeding Bird Survey species (Nebraska map of North America), and the Nebraska Department of Game and Parks who provided the Nebraska Breeding Bird Survey species (Nebraska map of North America).

1. North Carolina State University, Forestry Department, Raleigh, NC
2. University of Maine, Department of Wildlife Ecology, Orono, ME
3. North Carolina State University, Statistics Department, Raleigh, NC



A bit text heavy
but not too bad.



Determining the Wear Resistance of Occlusal Splints in a Prospective Clinical Study

P. Ottl, P. Schmelz, A. Piwowarczyk, H.-Ch. Lauer

Dept. of Prosthodontics, School of Dentistry (Director: Prof. Dr. H.-Ch. Lauer), ZZMK (Carolaria), J. W. Goethe University, Frankfurt, Germany

Objective

- To determine quantitatively the wear resistance of a newly developed light-curing splint resin over a period in situ of six months.

Materials and Methods

Patients

n = 20 consecutive patients
(mean age: 34.7 years; 12 F, 8 M)

Inclusion criteria

- Natural dentition/fixed denture
- Complete dentition to at least the 1st molar and

for the stabilization splint sample:

- Insufficient occlusal support
- Increased occlusal loss of dental hard tissue

for the distraction splint sample:

- TMJ pain and
- Complete anterior dislocation of the disk without reduction with terminal reduction
- TMJ osteoarthritis



Fig. 1: Stabilization splint in situ

Resin splint material (Fig. 1)

- Light-curing (400–500 nm) resin made of high-molecular dimethacrylates with organic and inorganic fillers
- Does not contain methyl methacrylate

Study design

- Duration: 6 months
- Types of splints (maxilla, n = 10 each): stabilization splints, distraction splints
- Splint wear mode: 24 hours
- Examinations: before insertion (BI), at 4 weeks (4W), at 3 months (3M), at 6 months (6M)
- Occlusal adjustments were restricted to the time before 4W.

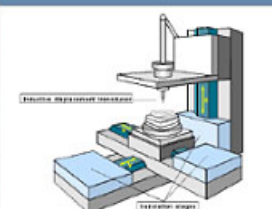


Fig. 2: Test setup

Measuring technology (Fig. 2)

- Vibration-isolated table framework
- 3 translation stages (for directions x, y, and z) (DC-Motor) (PI, Waldborn)
- DV 4 stereomicroscope (Zeiss, Oberkochen)
- WA 20 inductive displacement transducer/ Spider8 digital 8-channel measurement unit/ Calman 32 software V2.1 (HBM, Darmstadt)
- Local coordinate storage for occlusal contacts during baseline measurements
- Ten measurements each in regions 13, 23, 16, 26 (BI, 4W, 3M, 6M)
- Splint repositioned on remount cast

Results

- The medians of the occlusal vertical gaps/losses (wear, resin loss, water sorption, etc.) are shown in Fig. 3 (stabilization splints) and Fig. 4 (distraction splints).



Fig. 3: Occlusal vertical gaps/losses (mm) of the stabilization splint over a period in situ of six months (n = 10 stabilization splints)



Fig. 4: Occlusal vertical gaps/losses (mm) of the distraction splint over a period in situ of six months (n = 10 distraction splints)

- Statistical analysis (Mann-Whitney U-test, $p \leq 0.05$) showed no significant differences when comparing the corresponding results of stabilization and distraction splints.



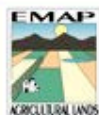
Figs. 5a and b: Lingual edge (maxilla) of the occlusal base with and without splint (Fig. 5a) and with distraction splint inserted (Fig. 5b) following six months of wearing.

Conclusions

- The present study clinically confirms the good wear resistance results of the new resin splint material obtained in a previous *in-vitro* study [OTTL et al., Dtsch Zahnärztl Z 52, 342 (1997)].
- Good wear resistance is of great importance for maintaining the therapeutic mandibular position during the treatment period (Figs. 5a and b).



Nice poster



A Framework for Assessing the Condition of Agricultural Lands

George Hess¹, Anne Hellkamp², Mike Munster³, Steve Peck³, Lee Campbell¹, Betty McQuaid⁴, Steve Shafer^{3,5}

Mission: To develop indicators of the condition of agricultural lands within an ecological framework, and to monitor and evaluate this condition on a regional basis.



Sustainable agriculture has been discussed, debated, and discussed in countless papers.

Definitions tend to be broad and encompass ecological, economic, social and even policy dimensions. Although these dimensions are universal, each may be measured independently.

In our efforts, we sought methods to examine only the ecological aspects of sustainability.

The ecological condition of agricultural land is defined by its productivity and the degree to which valued biotic and abiotic resources are conserved and protected.

Agricultural land in good condition is productive and does not compromise valued resources. **Sustainability** is the ability to maintain good condition over time.



Indicators were selected to reflect crop productivity and land stewardship.

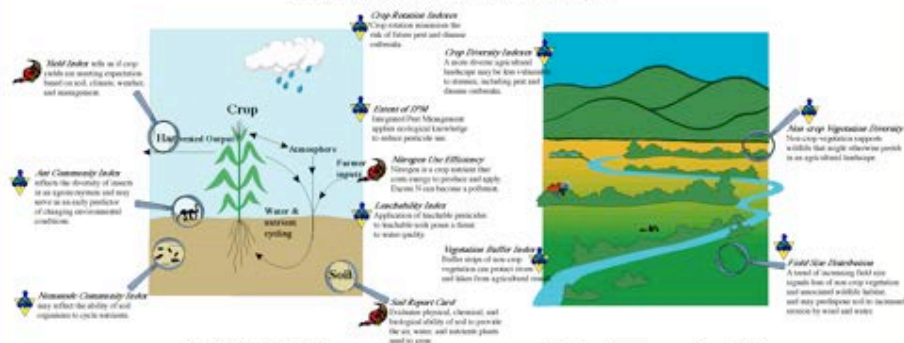
In making an assessment, condition is reported for each indicator. An overall condition may also be reported, but depends critically on the relative weighting of the goals for agricultural lands.

For sustainability, one can examine trends in crop productivity and stewardship practices.

Potential Indicators for Annually Harvested Herbaceous Cropland

As a starting point, we chose to concentrate our efforts on developing indicators for **annually harvested herbaceous cropland**—land planted with crops that are harvested every year whether the plants are annual or perennial. Common examples are corn, wheat, soybeans, alfalfa hay, and sorghum.

We also endeavored to supplement, rather than duplicate, existing efforts. Our conceptual framework is flexible enough to incorporate indicators based on data from other monitoring efforts. For example, an erosion indicator could be developed using the USDA Natural Resources Conservation Service's National Resource Inventory data.



Fields are for crops...

... but landscapes are for all of us.

Acknowledgements: The EMAP Agricultural Lands Research Group thanks the many individuals and organizations that made this effort a success. The individuals are too numerous to mention, but organizations include the USDA's Agricultural Research Service, Forest Service, National Agricultural Statistics Service, and Natural Resources Conservation Service; the U.S. Environmental Protection Agency; North Carolina State University; University of Maine; Oregon State University; University of Nebraska; and, with 1 grant, the list of organizations is pretty long, too. Thanks to all!

1. North Carolina State University, Forestry Department, Raleigh NC;
2. Duke University Medical Center, Durham NC;
3. North Carolina State University, Department of Plant Pathology, Raleigh NC;
4. USDA Natural Resources Conservation Service, Raleigh NC;
5. USDA Agricultural Research Service, Raleigh NC.



Where do I begin?

PREVALENCE OF OBESITY AMONG INNER CITY LATINO CHILDREN AND ADOLESCENTS

Nazrat M. Mirza MD, ScD, Jill Merchant MS, Leslie Beker, PhD

Children's National Medical Center and George Washington University School of Medicine and Health Sciences, Washington, DC

Background: Obesity is a multi-causal and public health problem facing children and adolescents in the U.S. Of particular significance is the increasing prevalence of obesity and its complications among the Latino population. Among this ethnic group there is a strong sense of family and children are a priority. Because of the pressures placed on children, they may be a misplaced expectation that children should not be denied food or other favorite meals at home. Obesity in children and adolescents is concerning not only because of the associated health and psychosocial consequences, but also because obese children tend to become obese adults. Since obesity is associated with being chronic disease, it will have no insurance impact on the healthcare system.

Purpose of Study: To estimate the extent of obesity among inner city Latino children and adolescents with the overall goal of assessing the need for an obesity intervention program.

Study Design: Cross-sectional and review. Six charts of children and adolescents aged 4 to 17 years were randomly selected from staff child visits to Children's Hospital's Latino Clinic, Ohio, for the calendar year 2006. The charts were an average of 80% to present a family, approximately 50% Latino, predominantly from El Salvador. Information extracted from the charts included height, weight, blood pressure, current classification, history, and physical findings consistent with obesity complications. Height (cm), weight (kg), BMI (kg/m²) was calculated from measured height and weight. Data analysis was done using SAS version 9.1.

Results: The distribution of the study sample is shown in Table 1. About 50% were females. The mean age was 10.4 years with a SD of 3.5 and a range of 4.0 to 16.7 years. The mean BMI was 20.8 with a SD of 3.4 and a range of 13.1 to 31.6. Overall 40% of the children and youth were overweight (BMI ≥ 25 percentiles) or at risk for overweight (BMI ≥ 85th percentiles), with an almost equal distribution between the two categories. Table 2: Males were more overweight and at risk for overweight than females, but the gender difference was not statistically significant. The prevalence of overweight was highest for girls ages 10 to 17 years.

Table 1 - Population statistics

Variable	Frequency (%)
Gender	
Male	50.4
Female	49.6
Age Categories (years)	
4-5	16.7
6-8	22.4
9-10	27.4
11-12	19.4
13-15	10.4
16-17	11.8
18-19	3.2

Results continued: Table 3 shows the distribution of overweight and at risk for overweight by age category. There did show that prevalence overweight and at risk for overweight is high in children as young as 4 to 5 years. Although the prevalence of overweight and at risk for overweight was lower in the age group 13-17 years, the difference was not statistically significant. Patient from help 0.84 and 0.8100 respectively.

Anthropometry was higher among the overweight than the non-overweight children and youth (p < 0.001). Patient from help 0.84. There was no difference in the frequency of occurrence of other signs such as hypertension, diabetes, asthma, sleep apnea, learning difficulties, behavior problems, osteoporosis, and ADHD between the overweight and non-overweight group. Only 7% of all the overweight children had their cholesterol levels checked. The cholesterol levels ranged from 132 to 300 mg/dL. The percent of the children had their serum triglyceride checked, and the range was 17 to 177 mg/dL. There was no significant association between overweight and either or cholesterol level or triglyceride in this study sample. Only 20% of the overweight children and youth were diagnosed and treated for their obesity according to their health care providers. There were no referrals for overweight intervention and other actions.

Table 3 - BMI distribution

BMI Category	Frequency (%)
At Risk for overweight (BMI 85-94 th)	
1. Male (n=125)	20.8
2. Male (n=70)	22.4
3. Female (n=47)	19.4
Overweight (BMI ≥ 95 th Percentile)	
1. Male (n=125)	22.4
2. Male (n=70)	18.1
3. Female (n=47)	20.2

Table 4 - At Risk for Overweight and Overweight by Age Category

Age Category (n)	At Risk for Overweight (%) (BMI 85-94 th)	Overweight (%) (BMI ≥ 95 th)
4-5 (n=16)	31.0	18.8
6-8 (n=22)	36.4	22.4
9-10 (n=27)	9.1	18.2
11-12 (n=19)	52.6	27.8
13-15 (n=19)	26.3	21.1
16-17 (n=10)	20.0	30.0
18-19 (n=3)	33.3	0.0

Conclusions & Recommendations: The prevalence rate for overweight and at risk for overweight among children and youth in this inner city Latino community is more than twice the national average. Primary health care providers need to acknowledge and accept the presence of obesity and overweight in children and adolescents to study and provide appropriate management of the problem. Targeted intervention and prevention strategies for overweight and obesity in children and adolescents are urgently needed for this population.



I'm feeling sleepy

Early Outcomes of the First 1471 Consecutive Kyphoplasty Procedures in the United States for the Fixation of Painful Osteopenic Vertebral Body Compression Fractures (VCF)

Steven R. Garfin¹, M.D., leader; R. Lieberman², M.D., Mark A. Bailey³, M.D., Joseph M. Lane⁴, M.D., Frank W. Phillips⁵, M.D., Hallett B. Mathews⁶, M.D., Hansen A. Yuan⁷, M.D., Barton H. Sachs⁸, M.D., for the Kyphoplasty Study Group
¹University of California, San Diego, Medical Center, San Diego, CA; ²Cleveland Clinic, Cleveland, OH; ³Berkley Orthopedic Medical Group, Berkley, CA; ⁴Hospital for Special Surgery, New York, NY; ⁵University of Chicago Spine Center, Chicago, IL; ⁶Mit-Memorial Spine Specialists, Richmond, VA; ⁷State University of New York Health Sciences Center, Syracuse, NY; ⁸Wiley Medical Center, Albany, NY

BACKGROUND

- 700,000 VCF's per year
- 275,000 diagnosed, >80% due to pain
- Spinal deformity associated with
 - Significant morbidity
 - 23% increased mortality (Kado, Ann Int Med 1999)
- Current treatments ineffective
 - Open surgeries fail
 - Medical management palliative
- Vertebroplasty
 - Bilateral transpedicular cement fill
 - Relieves pain
 - Requires high pressure and runny cement
 - High risk of cement leaks
 - Up to 73% where documented (Hest et al., Radiology 1997)
 - Major complications (Chen, J Int Neuror 1997)
 - 1.3% in retroperitoneal
 - 10% in metastatic cancers

KYPHOPLASTY

Kyphoplasty is a minimally invasive orthopedic procedure for reducing and fixing painful vertebral body compression fractures secondary to osteoporosis. Using a posterior approach, one or two inflatable Bone Tamps (Fig. 1) are inserted into the fractured vertebral body, generally using a bilateral transpedicular approach (Fig. 2). The surgeon carefully inflates the balloon tamps (Fig. 2) using radiopaque contrast medium with image, volume and pressure control. The increased balloon tamp volume compacts the inner cancellous bone as it pushes the fractured outer cortical bone back toward its normal position. The inflation path is also controlled by placement, volume and balloon design. After reduction, the balloon tamp is removed, and the resulting void is filled with thick PMMA under low manual control and low pressure. The steps of Kyphoplasty are illustrated in Fig. 3.

Figure 1: Inflatable Bone Tamp (IBT)



Fig. 1



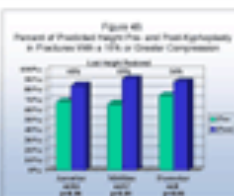
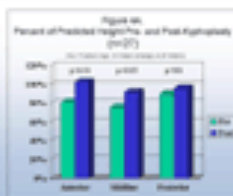
Fig. 2



Fig. 3

STUDY DESIGN AND METHODS

A retrospective multi-center review to assess early outcomes with Kyphoplasty. Pain was localized by physical examination. The presence of marrow edema and collapse was confirmed on MRI. General or deep local anesthesia was chosen based on anatomy, number of levels and patient status. The first 125 patients at our centers were asked to characterize their back pain as improved, the same or worse 24 hours post-op and at last follow-up. Fractured and nearest normal vertebral body heights were measured anterior, middle and posterior in the first 27 vertebral body fractures treated by one surgeon (MAM). The height of the nearest normal vertebral body was used to calculate the % of predicted height for all the vertebral bodies (Fig. 4A) and for the sub-set where which had lost 10% or more of height before treatment (Fig. 4B).



The pre-treatment height was subtracted from the predicted height, then divided by the post-treatment height subtracted from the predicted height, to find the percentage of lost height restored. One set of X-rays by one surgeon (JMP) are used to show an example height restoration (Fig. 5A) and deformity correction (Fig. 5B). Device-related major complications from all procedures are reported. Phlebotomy leaks in the first 70 procedures performed by one surgeon (JMP) were assessed with X-ray and MRI.

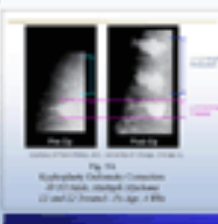


Fig. 5A

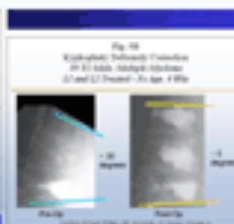


Fig. 5B

PRELIMINARY RESULTS

- 1471 fractures (single Table 1)
- Average fracture age: 45 months
- Range: 10 days to 15 years
- 90% operated
- 90% operated (single Table 2)
- Average fracture levels: 11 (range: 8-12)
- Average tamp inflation volume: 1 cc (range: 0.5-1.5 cc)
- Were the 90% operated patients:
 - Male/female
 - 80% report pain improvement at 1 month
 - 80% 10% reduction of fracture height (Fig. 6A, B, 10, 15)
 - No increased incidence of adjacent fractures
 - 10% device-related major complications
 - 4 neurologic
 - 1 pulmonary
 - 1 bleeding
 - 1 phlebotomy
 - 0% lost rate (during follow-up)

CONCLUSIONS

Kyphoplasty is an important treatment option that provides immediate mobility and return to activities of daily living to patients with acutely painful vertebral body compression fractures secondary to osteoporosis. Kyphoplasty facilitates fracture reduction and deformity correction. While reduction is more likely in acute fractures (few months or less), it has been seen in fractures over one year old. Kyphoplasty also provides rapid pain relief in the nearly all patients, and this result is independent of fracture reduction. The safety profile of Kyphoplasty compares favorably to the published safety profile of vertebroplasty.



OK, but
which way
do I go?



Poster title goes here, containing strictly only the essential number of words...



Author's Name/s Goes Here, Author's Name/s Goes Here, Author's Name/s Goes Here
Address/es Goes Here, Address/es Goes Here, Address/es Goes Here

Introduction

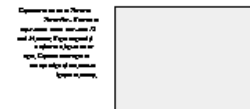
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Check with conference organisers on their specifications of board dimensions before you print your poster (e.g. most print poster sizes landscape portrait or square).
The page size of this poster template is A0 (36" x 108"), landscape (horizontal) format. Don't change the page size, MU can scale it to a smaller or larger size when printing. You need a different shape poster with either a portrait (vertical) or a square poster template.
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Tip for making a successful poster ...
• Re-write your paper in poster format. Simply over-tying and over-writing.
• Headings often than sentences should be both upper and lower case initial capitals. bold characters in text.
• Headings should be in capital letters or underlined to stress your points.
• When laying out your poster leave breathing space around your text. Don't overcrowd your poster.
• Try using photographs or colour graphics. Avoid using numerical tables.
• Split text and images into separate sections.



Experiments are run in the same way as in the previous poster. Experiments are run in the same way as in the previous poster. Experiments are run in the same way as in the previous poster.

Results

Importing the results ...
Images such as photographs, graphs, diagrams, logos, etc. can be added to the poster.
To insert a colour image into your poster go through the menu as follows: Insert > Picture > From File. In the 'Find my computer' dialog box, select the image file. The image will be inserted into the poster.
Be aware of the image size when importing. The average colour photo (13" x 18" at 300dpi) would be about 3MB. (1MB is 1024KB). Call MU if unsure.
Do not use images from the web.

How to add graphs ...
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Conclusion

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Acknowledgements

Just highlight the text and replace with your own text. Replace it with your text.



Perfect!



A Large-Scale Public Library Renovation in Taiwan

Library Association of E. G. C.
National Teachers' Library of the

ABSTRACT

There are 535 public libraries, including 100 urban, county, and village public libraries, in Taiwan. As most of them were built in 1950s, they are not fit in the digital environment to meet users' needs.

In order to upgrade the quality of public library services in Taiwan to meet users' needs and to attract library borrowing, in 2000, the central government of Taiwan approved a budget of NT\$1.2 billion (US\$ 4 million) as a long-term public library renovation project in 301 public libraries.

Midland Technical Library was designated as coordinate library to maintain the project from February 2003 to June 2004. 301 public libraries were divided into eight groups according to the geographical area, and a meeting committee was formed, consisting of committee members from the Public Library and Information Science, Architecture, Space Design, Literature, and History. 60 committee members were assigned to one of eight groups of 301 public libraries to help and to give suggestions of information, management, improvement, service programs of each library.

The project was conceived and completed efficiently and effectively in June 2004. This poster presentation will display the results of the successful, innovative, replicable, library management and services of 201 public libraries in Taiwan. The contents of the poster will be explained by words, pictures, and statistical tables.

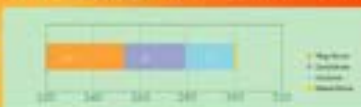
Expanded Public Libraries
<http://www.plf.org.tw>



Keywords: *Self-esteem, self-esteem threat, self-esteem threat sensitivity, self-esteem threat sensitivity scale, self-esteem threat sensitivity scale-2*

Category	Subtotal Revenue	100 Revenue	100 Revenue	100 Revenue	100 Revenue	100
100	1	1	1	1	1	100

Figure 2: Members of Wiggins at each administrative level involved in the community project.



100



There is always the project



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● <http://www.pearsoned.com>

● **Investing and saving**

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* *For more information, see the sidebar on page 10.*



● **Investment and Savings** – **different** – **mutually** **exclusive**



● **Unabhängigkeit** – Abgrenzung gegenüber anderen Organisationen



● **Transfer** – the process of moving a patient from one level of care to another, such as from the hospital to a nursing home or from a nursing home to a rehabilitation center.



Systemic methods



Oh my gawd!

It is worth noting that in our models, the curve for each individual problem is steeper than the last. It may be that the use of verbal or pictorial notation compensated for the effects of nonverbal. Graduation may have provided a way to help the verbal part of the brain to connect concepts. It may have a knock-off effect on nonverbal or an association that is somewhere.

Nice flow,
but too
metallic



Thomas: the author is 28 now, 49

and family among the
-the nation, the



Gorgeous!

LESSONS LEARNED FROM AIRWAY PRESSURE RELEASE VENTILATION (APRV)

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Departments of Surgery¹ and Emergency Medicine², Philadelphia, PA USA

INTRODUCTION

Airway Pressure Release Ventilation (APRV) (a.k.a. BiPAP) has been previously demonstrated to be a useful modality to manage patients with acute lung injury (ALI) or the acute respiratory distress syndrome (ARDS). As this is a fundamentally different mode than conventional cyclic ventilation, we required a single institution's experience with APRV to determine safety, complication detection, and efficacy at resolving hypoxemia and hypercarbia.

METHODS

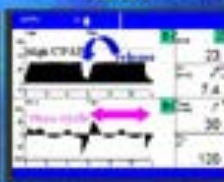
Consecutive patients transitioned from either volume or pressure targeted ventilation to APRV (Dräger Esch 4 Pulmonary Wocklation) at a University hospital surgical ICU were retrospectively reviewed. Patients initially ventilated with APRV were excluded. Initial APRV settings to correct hypoxemia ($pO_2 \leq 60$ torr or $FIO_2 \geq 0.9$) were a P_{high} at the prior plateau pressure, a T_{high} of 6.0 sec and a T_{low} of 0.8 sec. Hypercarbia ($pCO_2 \geq 55$ torr and $pH \leq 7.3$) patients were set at a T_{high} of 5.0 sec and a T_{low} of 1.0 sec. Settings were adjusted to resolve hypoxemia and hypercarbia. IRB approved abstracted data included principal diagnoses, ventilation parameters, laboratory values and ventilator associated complications. Data before and after APRV were compared using a two-tailed paired t-test or Chi-square as appropriate; significance was assumed for $p < 0.05$ (^{1,2}).

RESULTS

Demographics

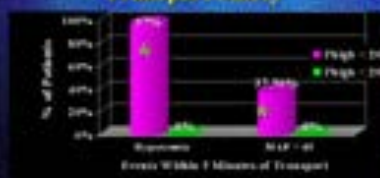


APRV

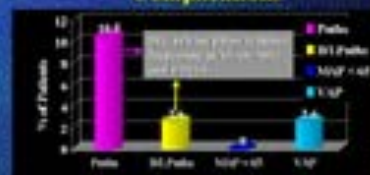


Element	Value
% Hypoxemia	88%
% Hypercarbia	12%
Time to $SpO_2 \geq 92\%$	7 ± 4 min
Time to $FIO_2 \leq 0.6$	5.2 ± 0.9 hr
Time to $pCO_2 \leq 40$ torr	42 ± 7 min
Time to max ΔpCO_2	76 ± 12 min
Mean change in V_E	-3.5 ± 0.9 L/min ³

Transport Safety



Complications



CONCLUSIONS

1. APRV is a safe rescue mode for hypoxemic or hypercarbic respiratory failure and requires a significantly lower V_E than conventional ventilation.
2. Decreasing release phase volumes and a rising pCO_2 are strong indicators of pneumothorax in a patient on APRV. Routine end-tidal CO_2 monitoring is recommended.
3. Preparation for safe intra-hospital transport may be keyed to the P_{high} required for oxygenation and ventilation. Patients requiring a $P_{high} > 20$ cm H_2O should be transported on the ventilator.



Welcome to
the 80's
Fer sure!



Helpful sites on poster presentations:

<http://colinpurrington.com/tips/academic/posterdesign>

<http://www.ncsu.edu/project/posters/NewSite/>

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